AI 500
Foundations of Artificial Intelligence

L13: Convolution Neural Network Part 2

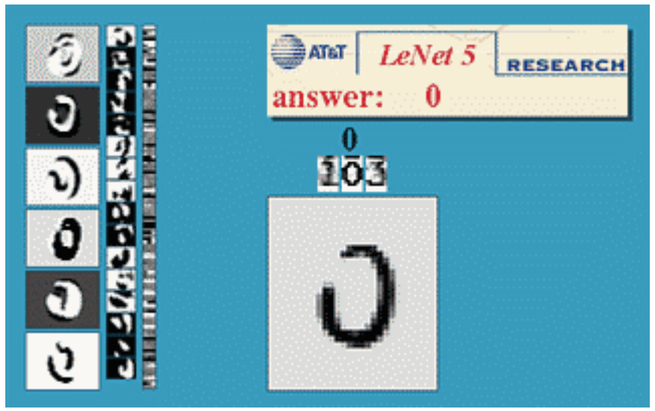
By Murtaza Taj

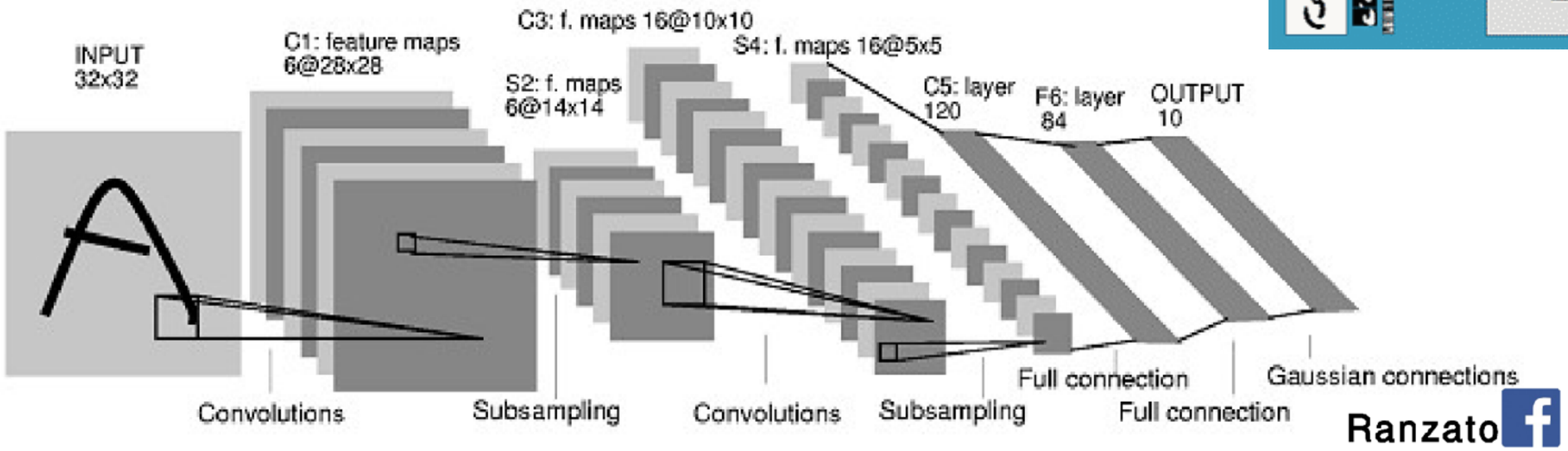
LUMS
A Not-for-Profit University



Convolutional Neural Network

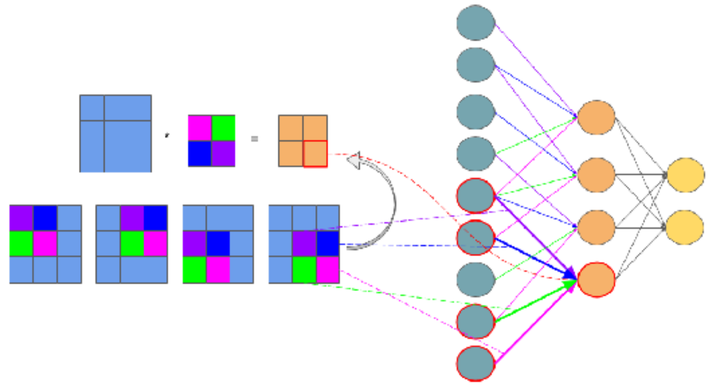
- Le Cun 1998



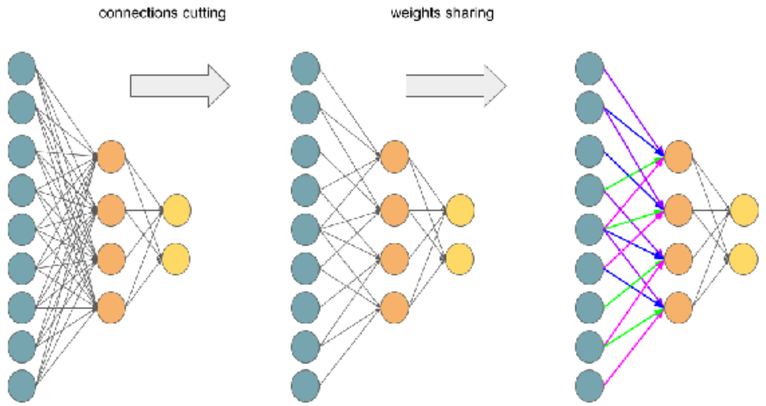


Agenda

- Forward and Backward Propagation using Convolution operation



- Transforming Multilayer Perceptron to Convolutional Neural Network



Recap - Convolution - Numerical Example



Image

4	7	7	5	0	8
1	8	5	7	8	4
2	8	4	3	1	0
0	7	4	7	8	3
4	2	0	0	6	8
3	6	0	3	1	4

Filter

1	0	-1
1	0	-1
1	0	-1

Ignoring flip

Conv Output

-9	8	7	3
-10	6	-4	10
-2	7	-7	-1
3	5	-11	-5

Recap - Convolution - Smoothing Filter

Foundations of AI AI500

Image

128	128	128	52	52	52
128	128	128	52	52	52
128	128	128	52	52	52
52	52	52	128	128	128
52	52	52	128	128	128
52	52	52	128	128	128

Filter

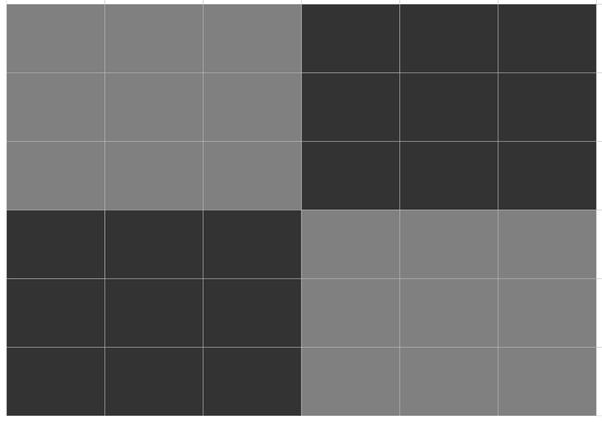
1	1	1
1	1	1
1	1	1

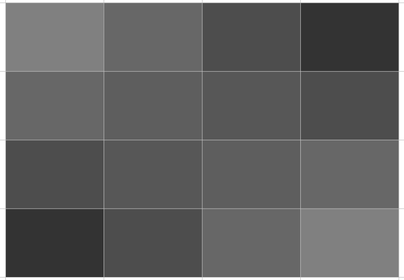
x1/9

Ignoring flip

Conv Output

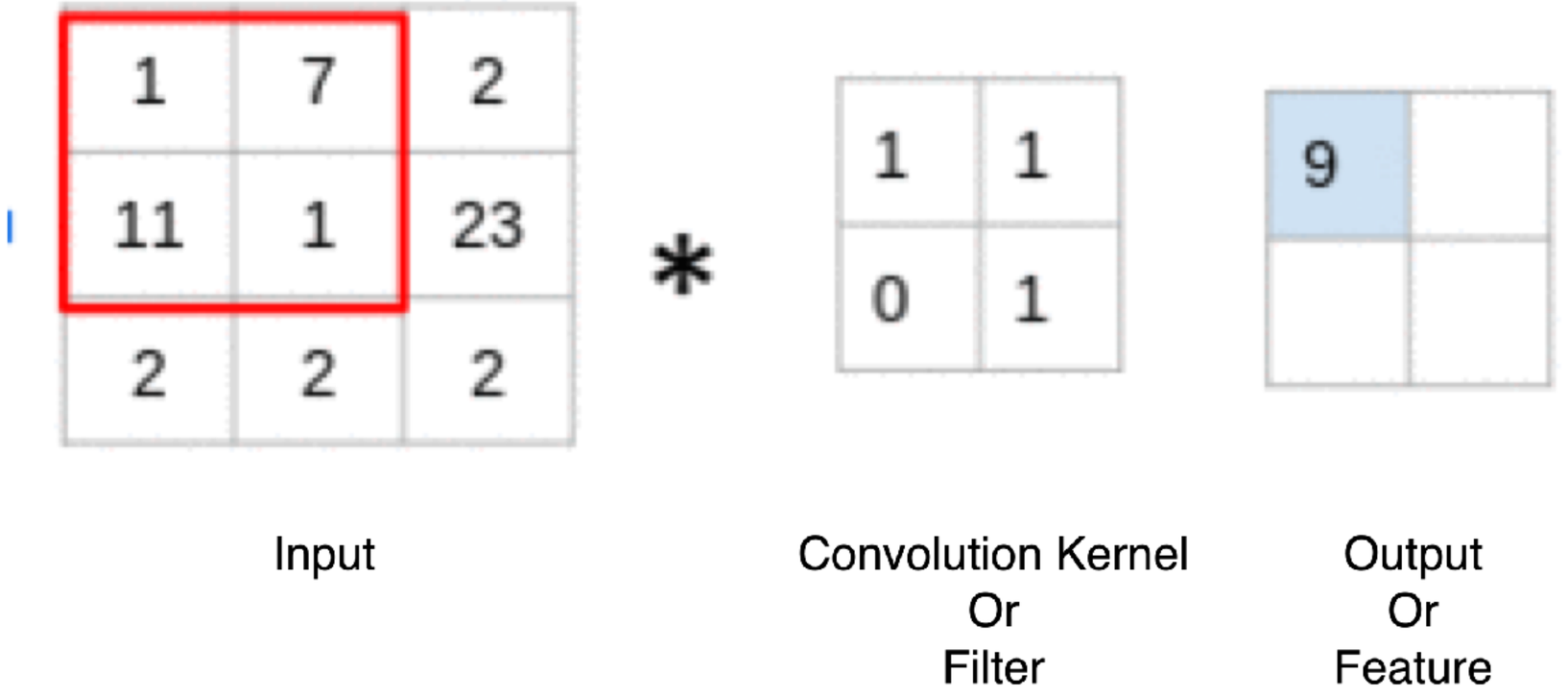
128	103	77	52
103	94	86	77
77	86	94	103
52	77	103	128



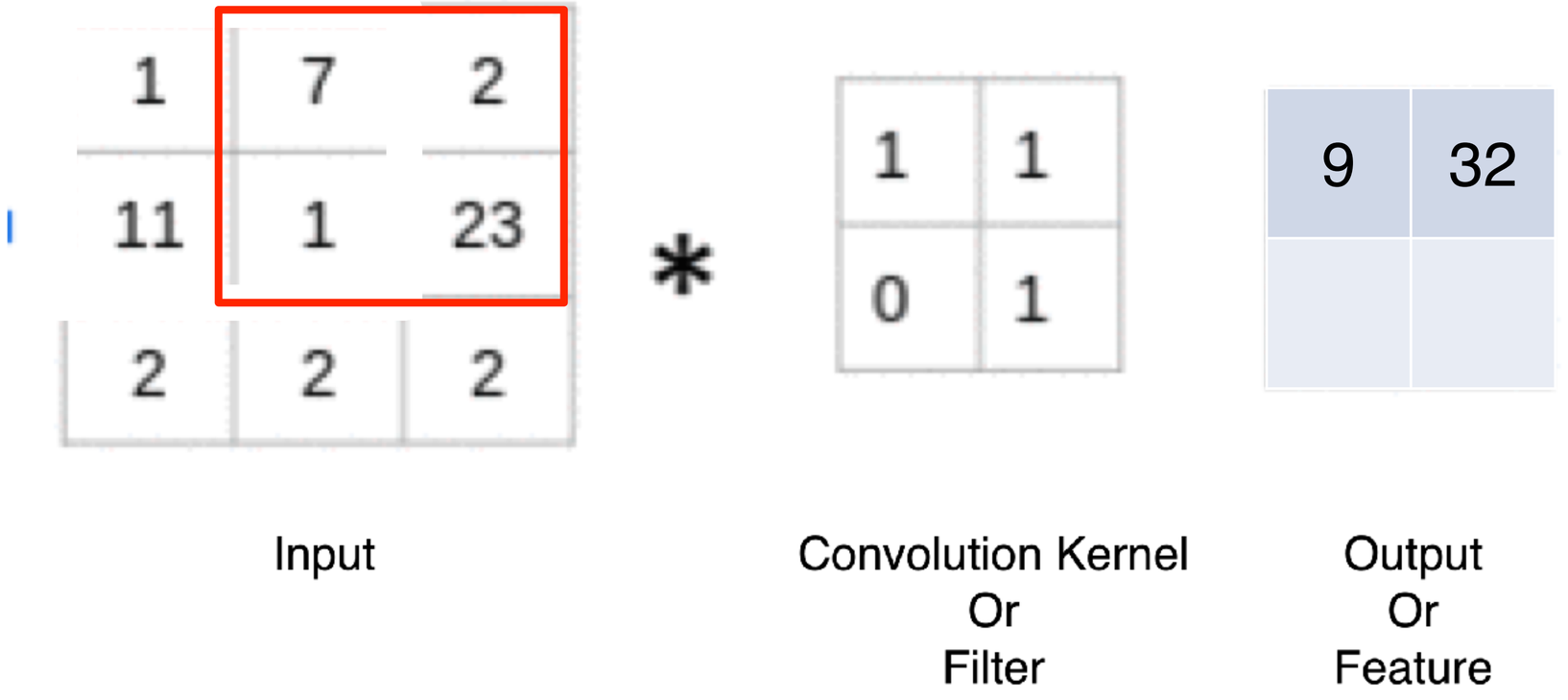


Transforming Multilayer Perceptron to Convolutional Neural Network

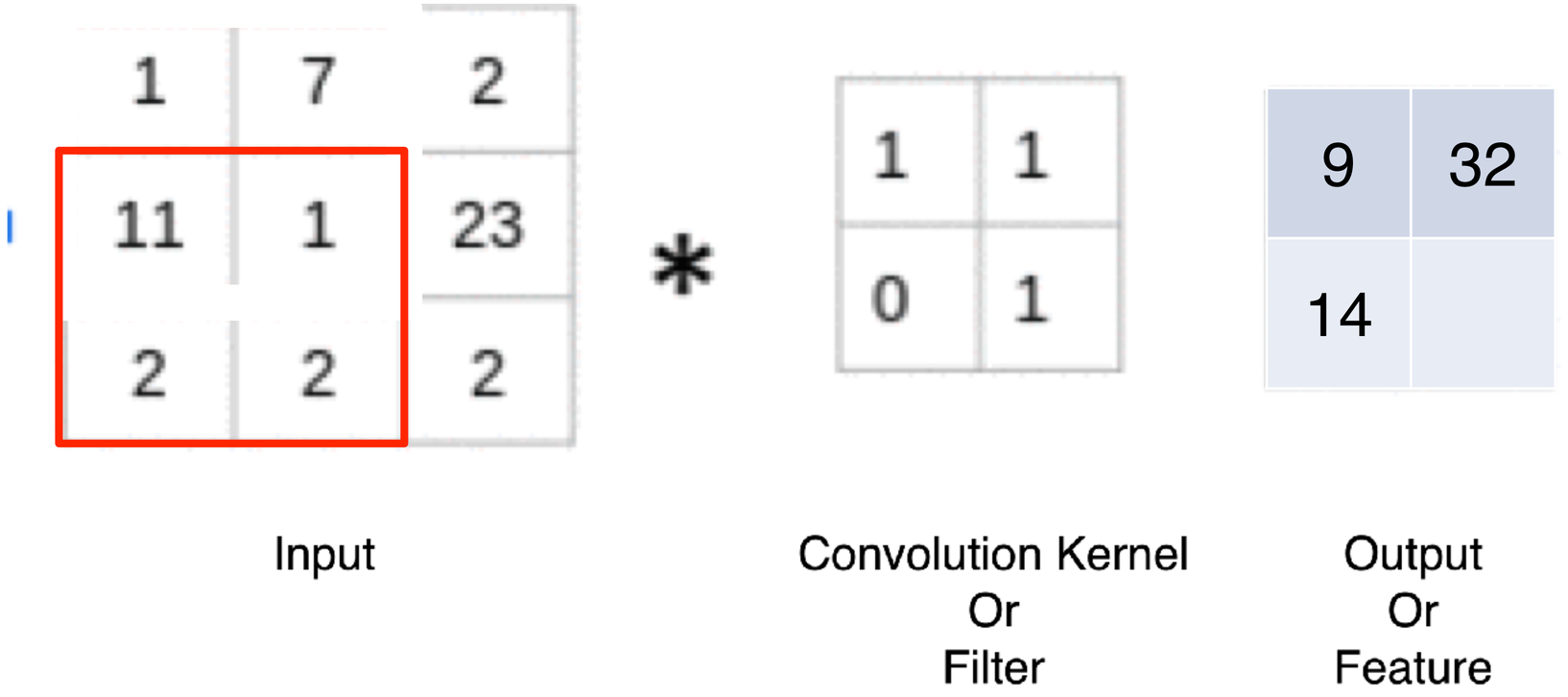
Recap: Convolution



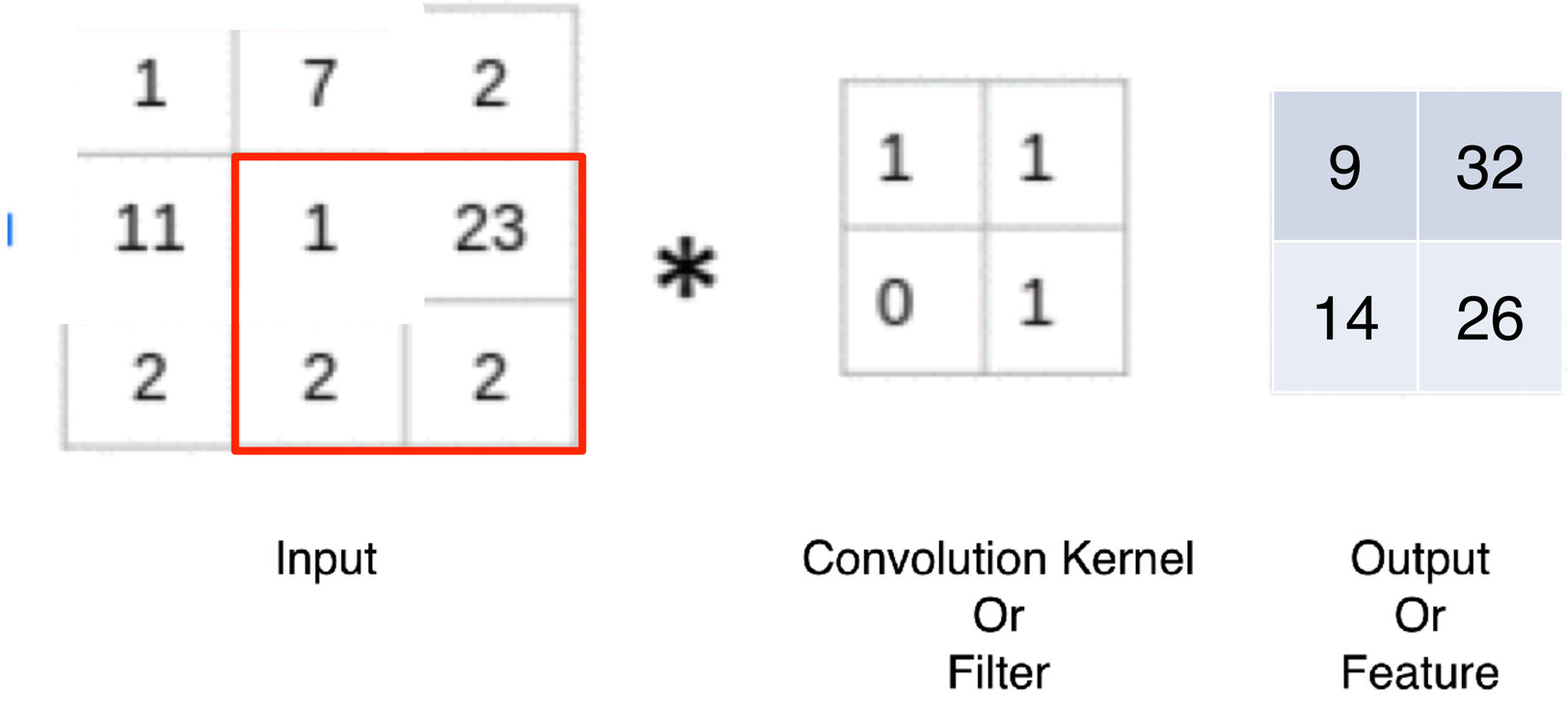
Recap: Convolution



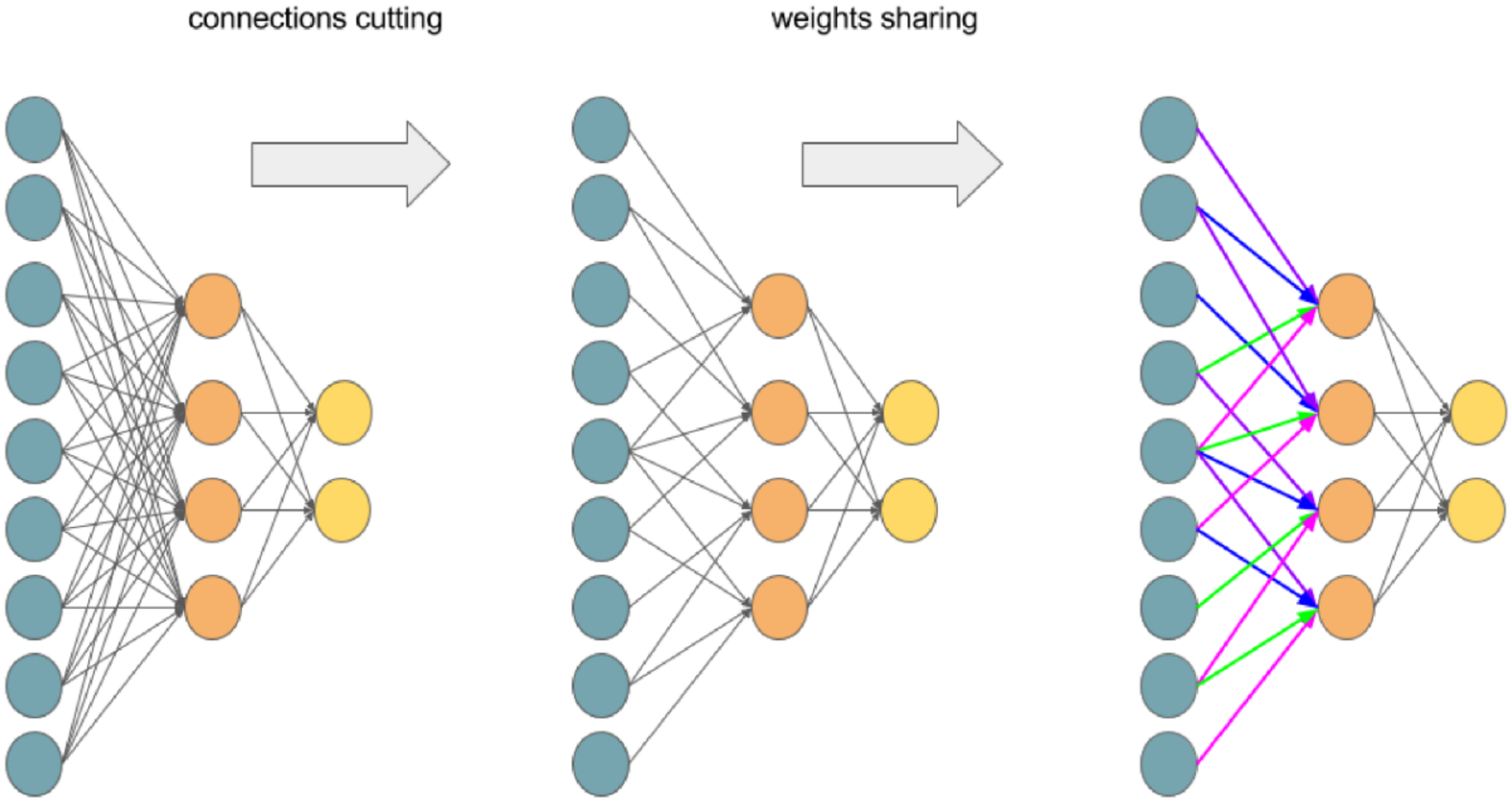
Recap: Convolution



Recap: Convolution

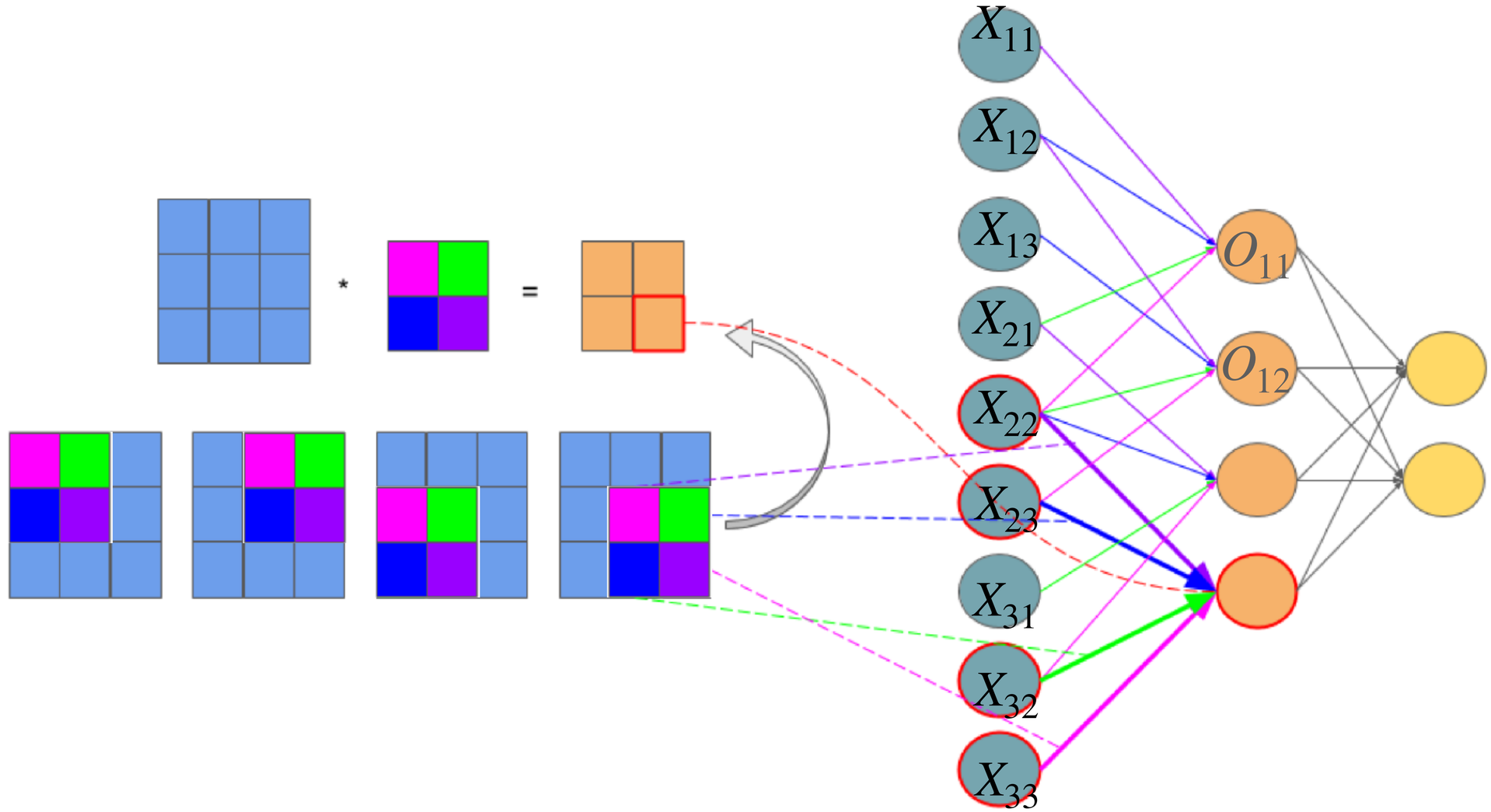


Transforming Multilayer Perceptron to CNN



https://grzegorzgwardys.wordpress.com/2016/04/22/8/

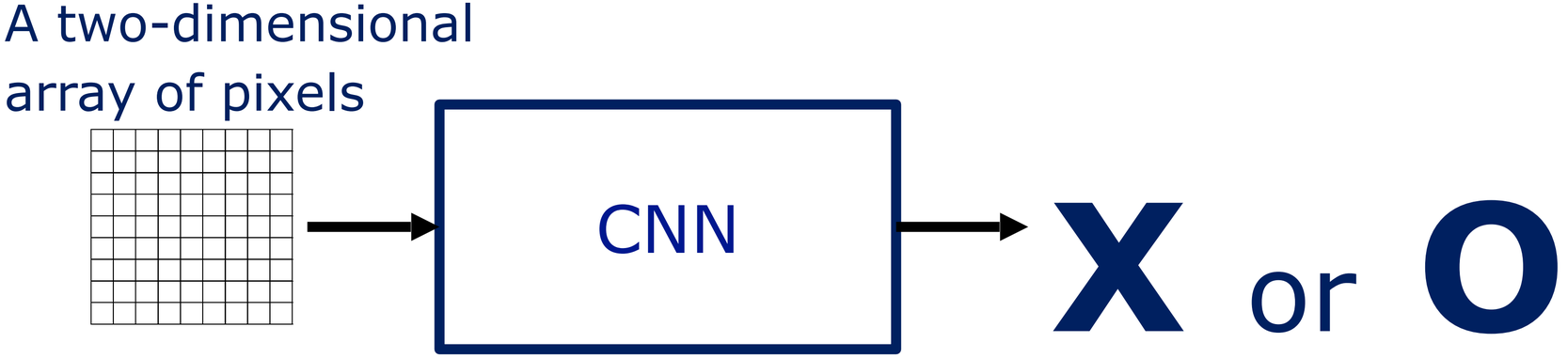
Feedforward in CNN is Identical with Convolution Operation



https://grzegorzgwardys.wordpress.com/2016/04/22/8/

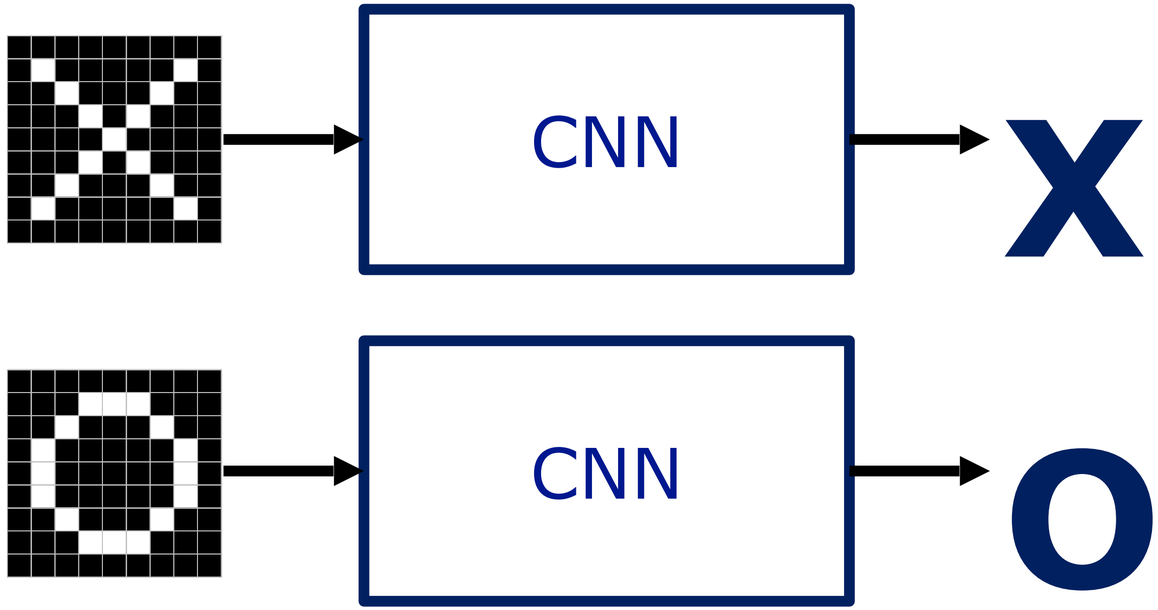
A toy ConvNet: X's and O's

Says whether a picture is of an X or an O



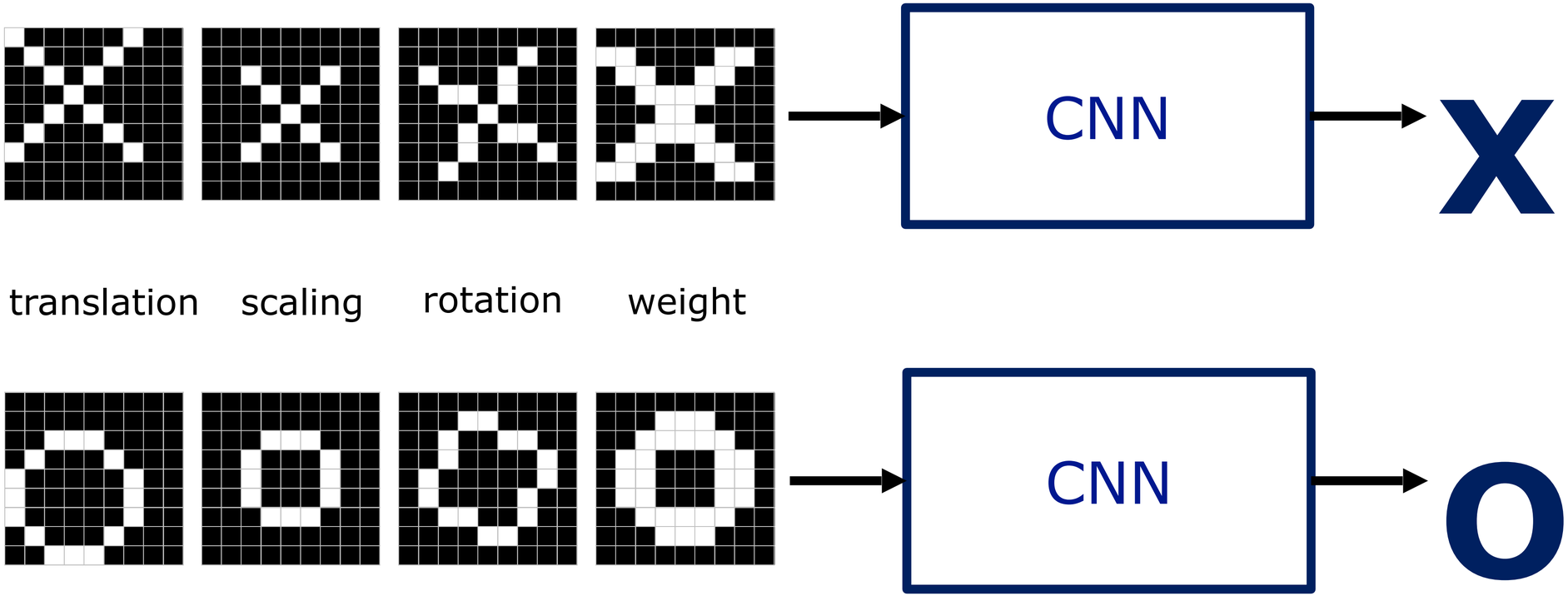
Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

For example



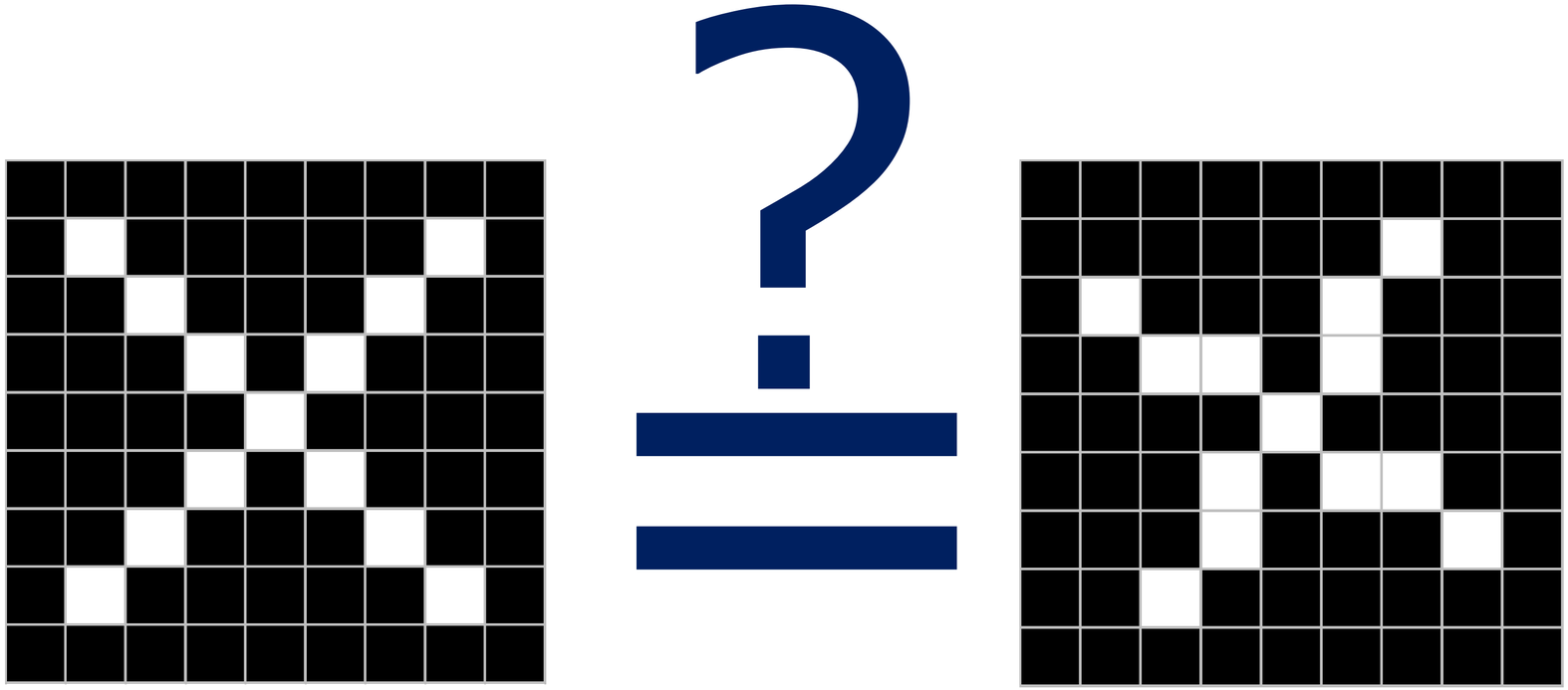
Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Trickier cases



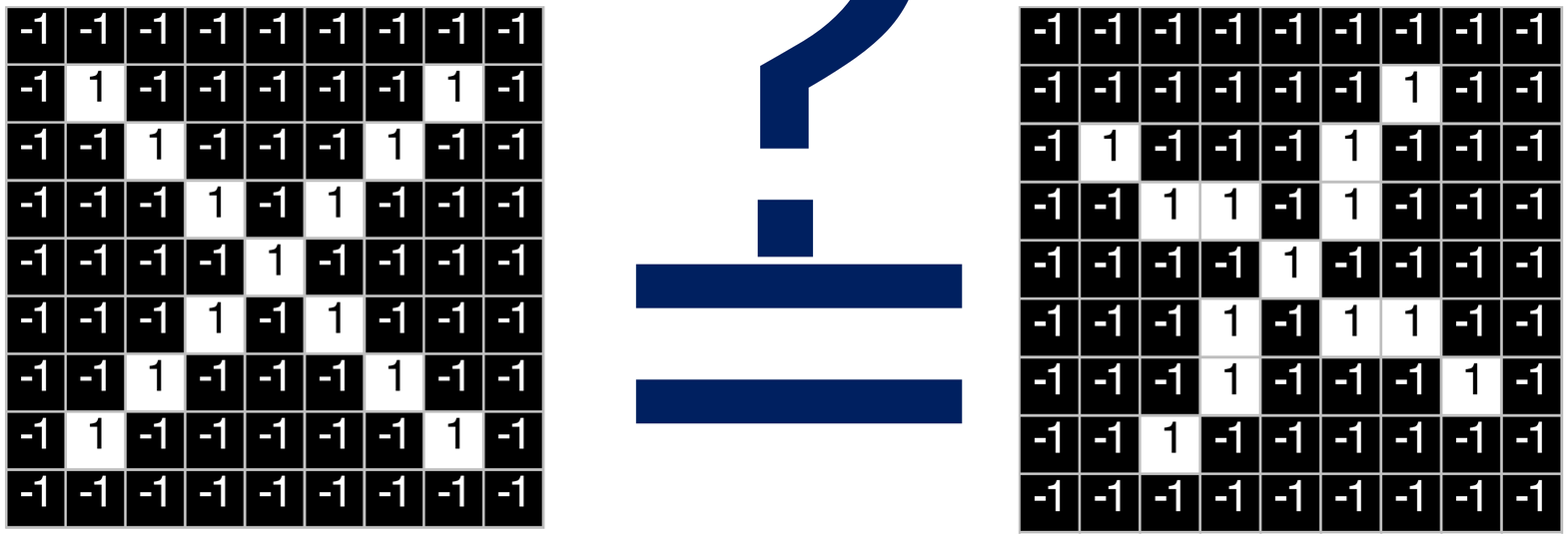
Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Deciding is hard



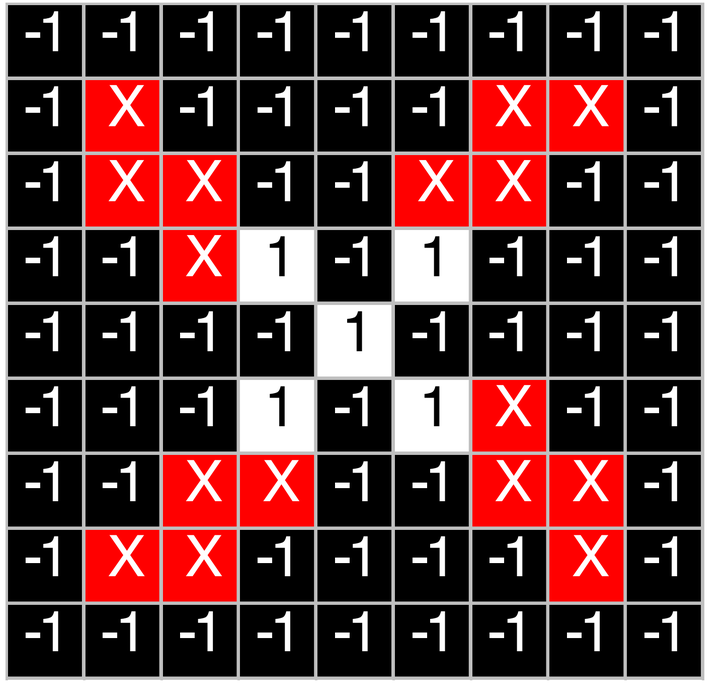
Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

What computers see



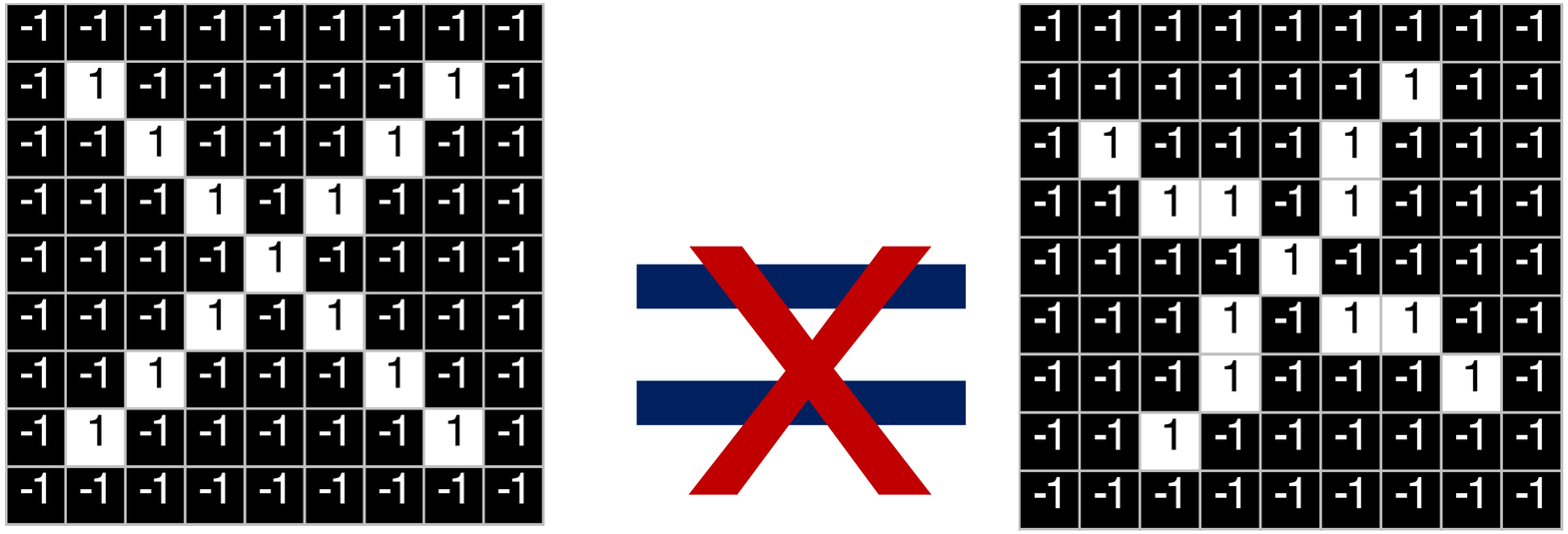
Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

What computers see



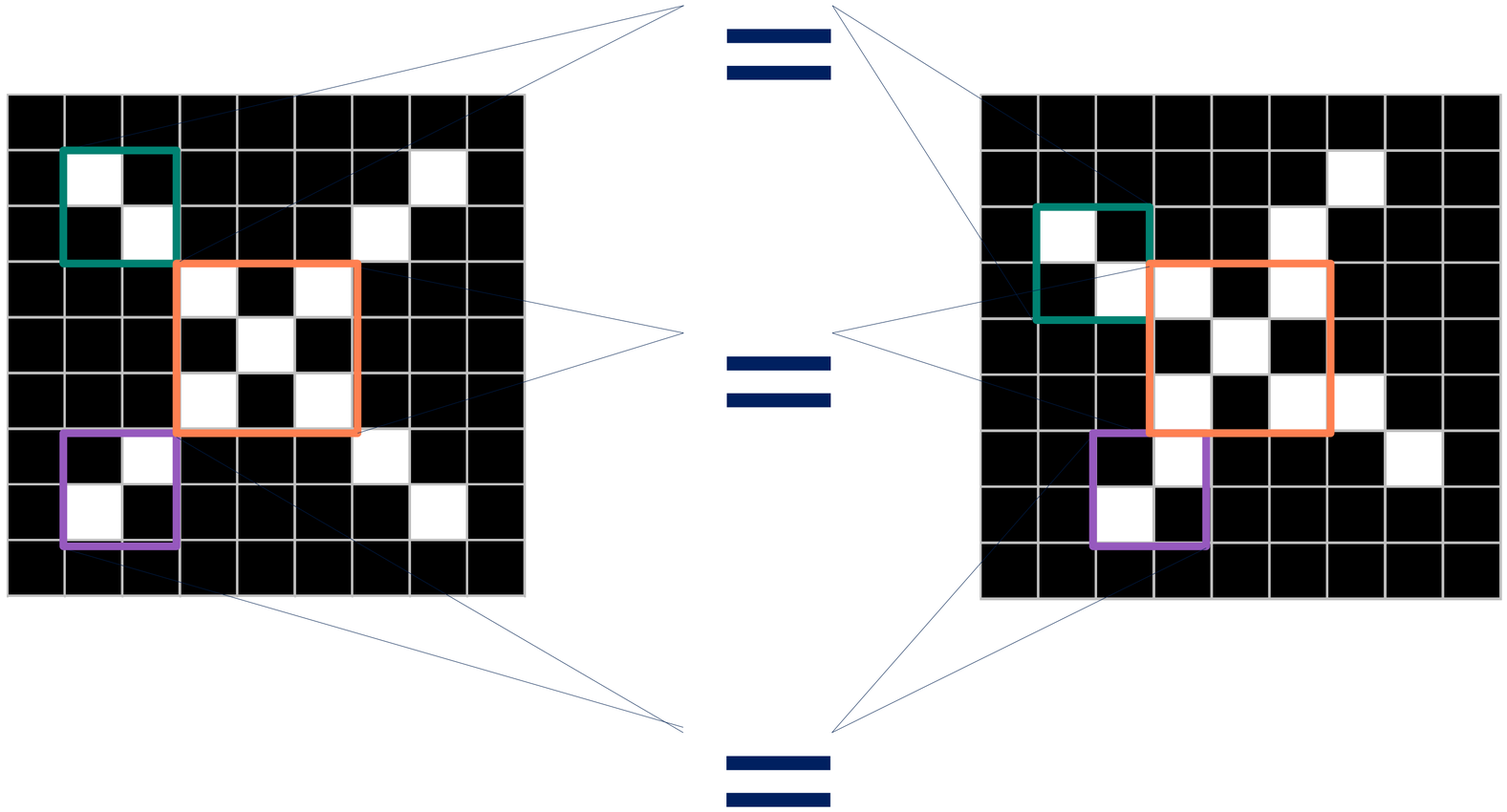
Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Computers are literal



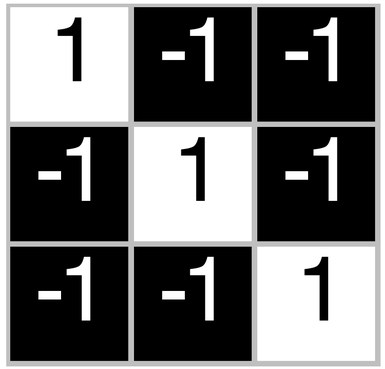
Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

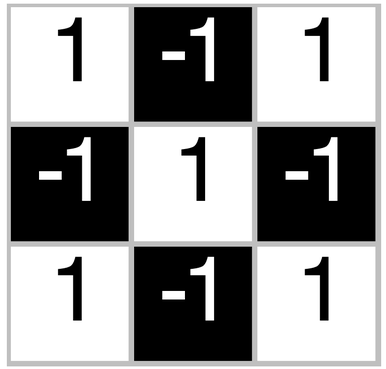
ConvNets match pieces of the image

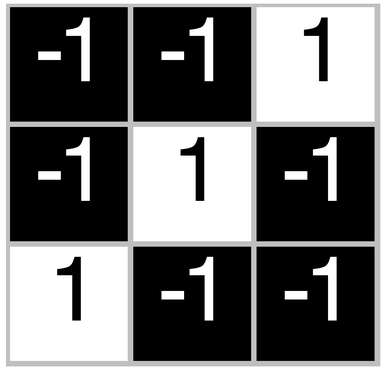


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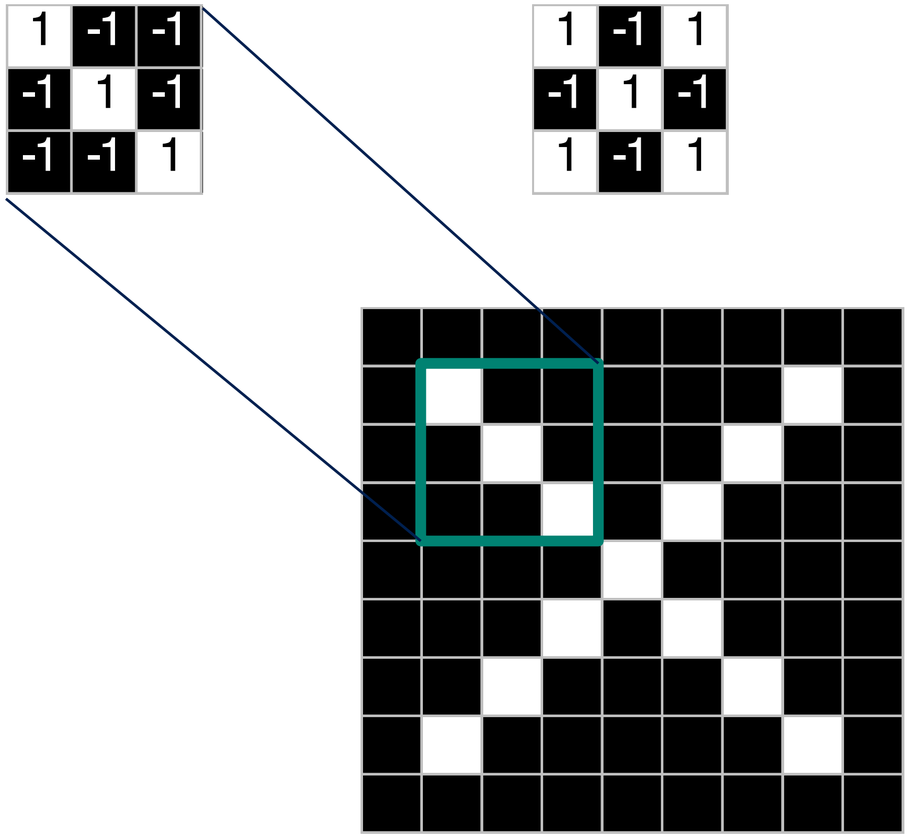
Features match pieces of the image



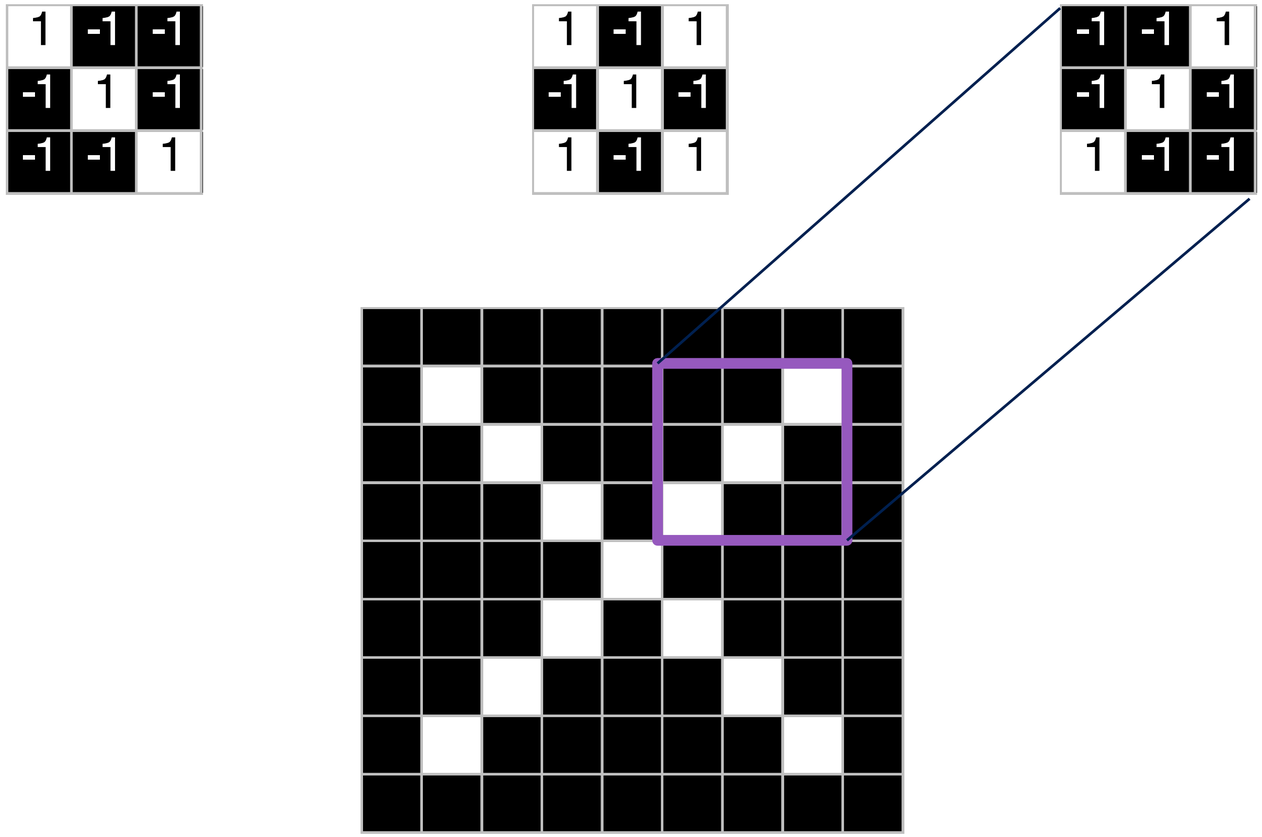




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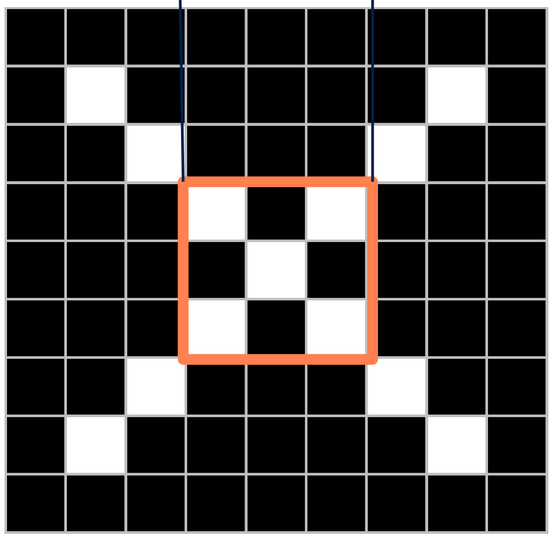


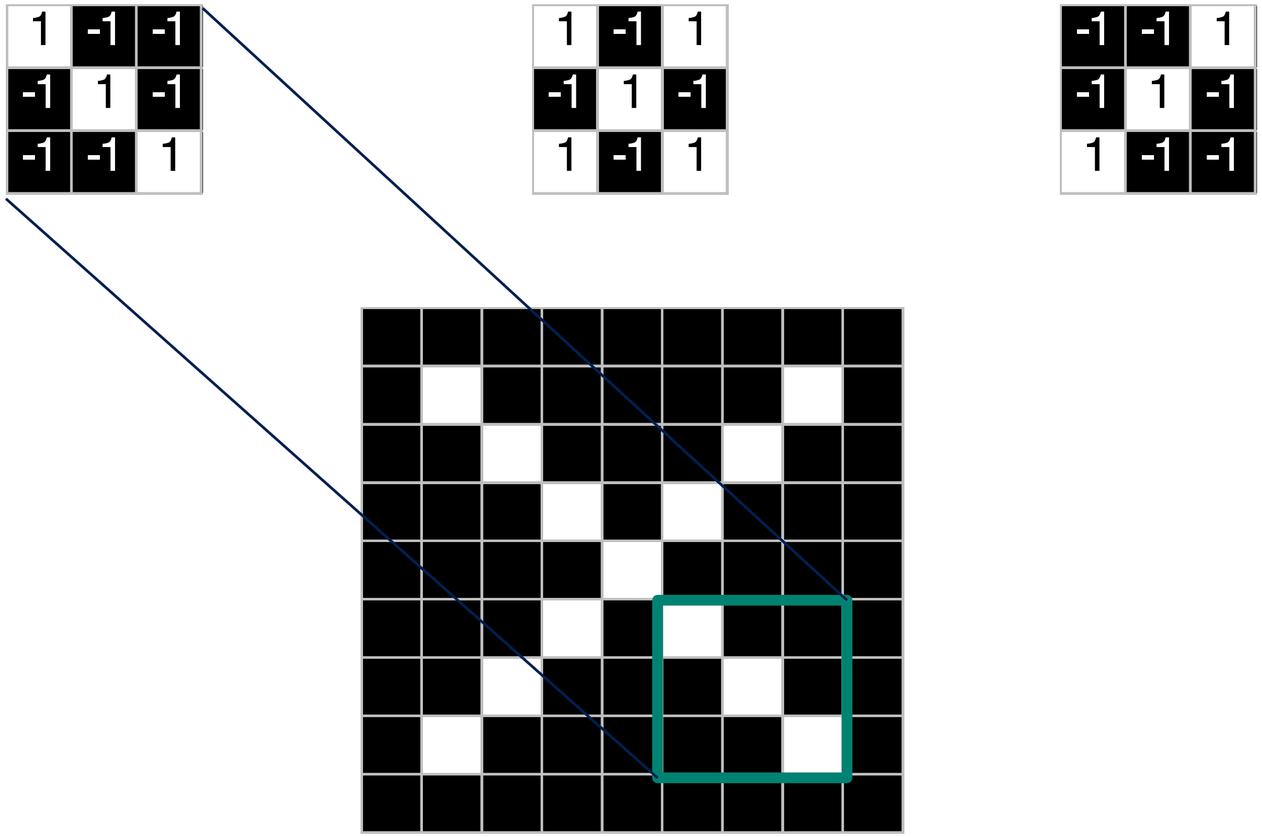
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1	-1	-1
-1	1	-1
-1	-1	1

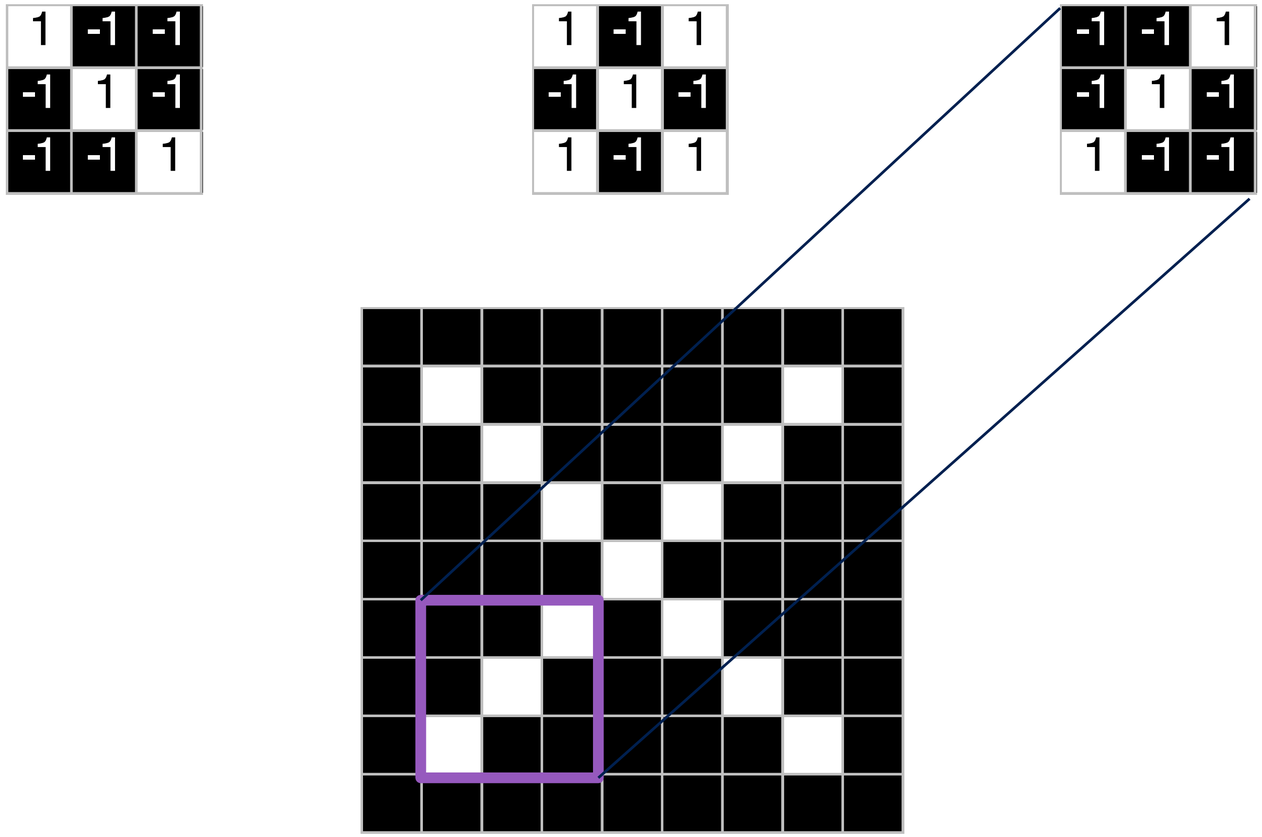
1	-1	1
-1	1	-1
1	-1	1

-1	-1	1
-1	1	-1
1	-1	-1



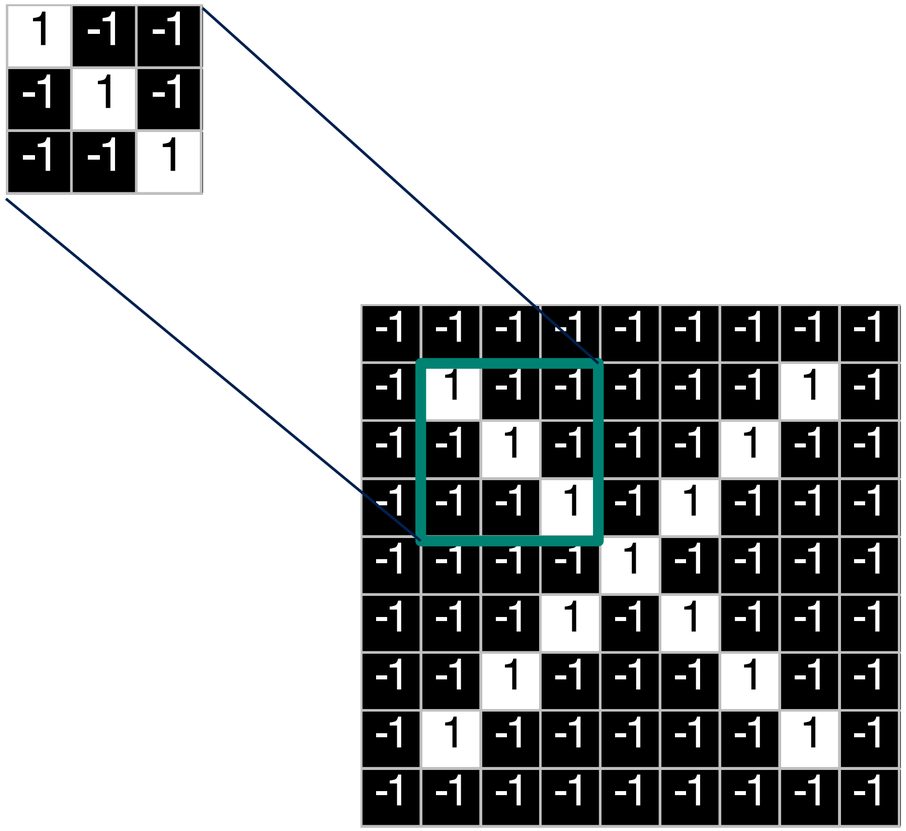


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Filtering: The math behind the match



Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

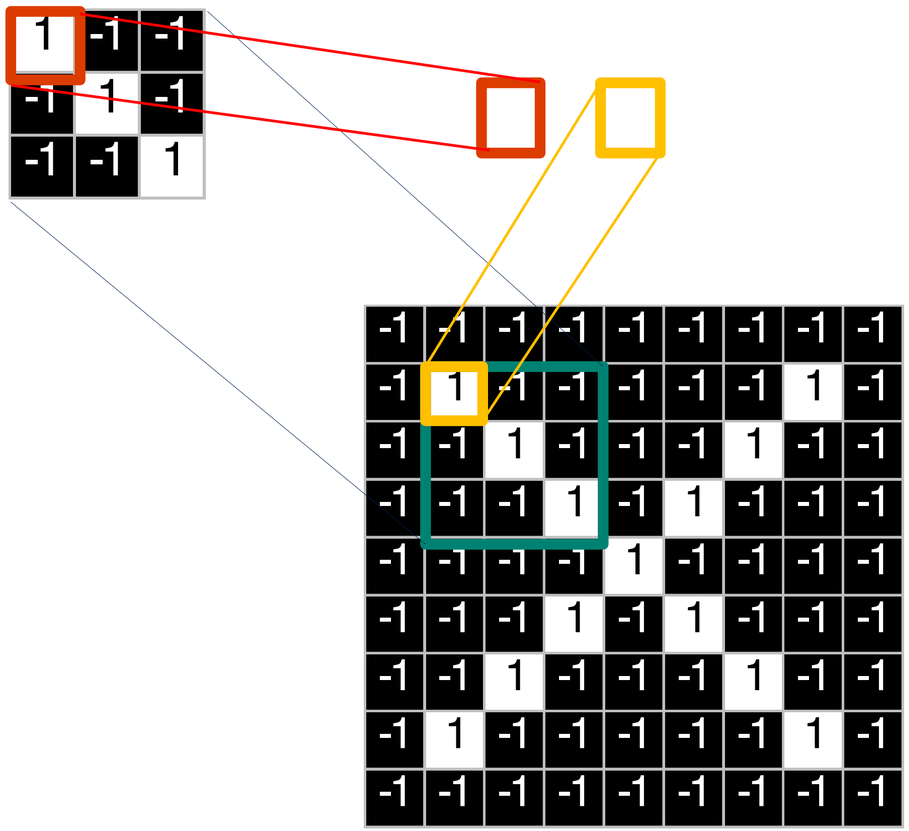
Filtering: The math behind the match



1. Line up the feature and the image patch.
2. Multiply each image pixel by the corresponding feature pixel.
3. Add them up.
4. Divide by the total number of pixels in the feature.

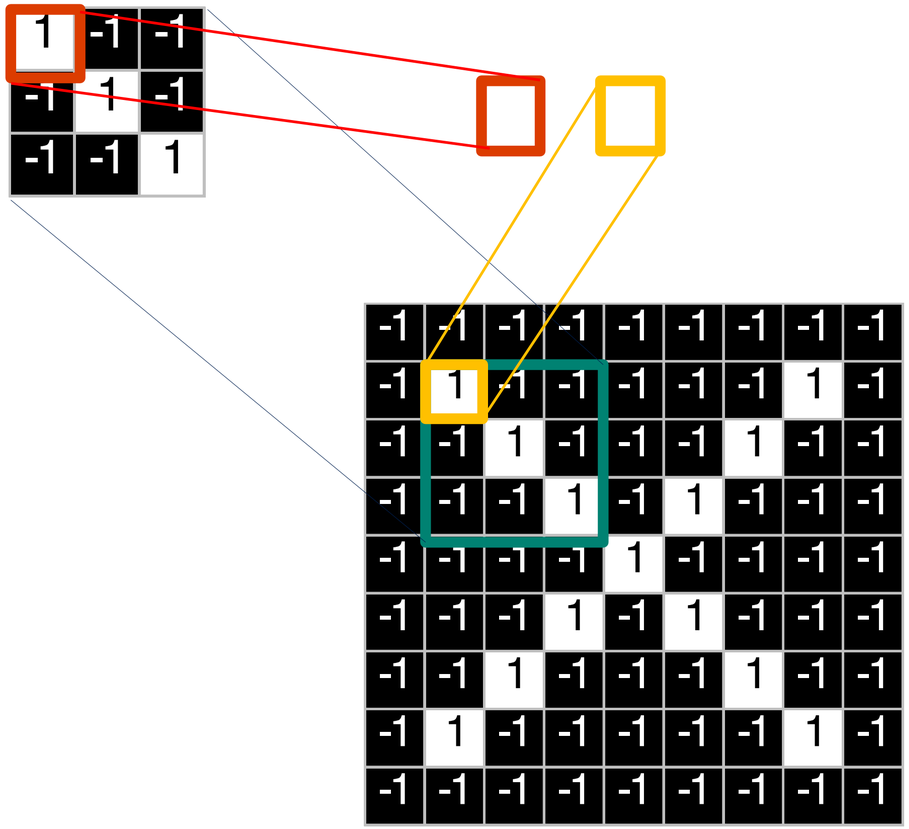
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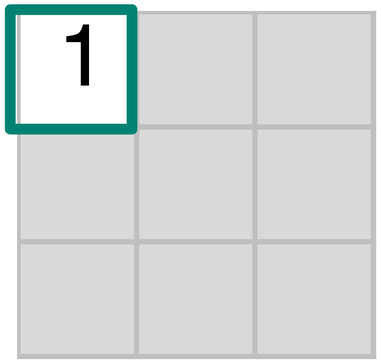
Filtering: The math behind the match



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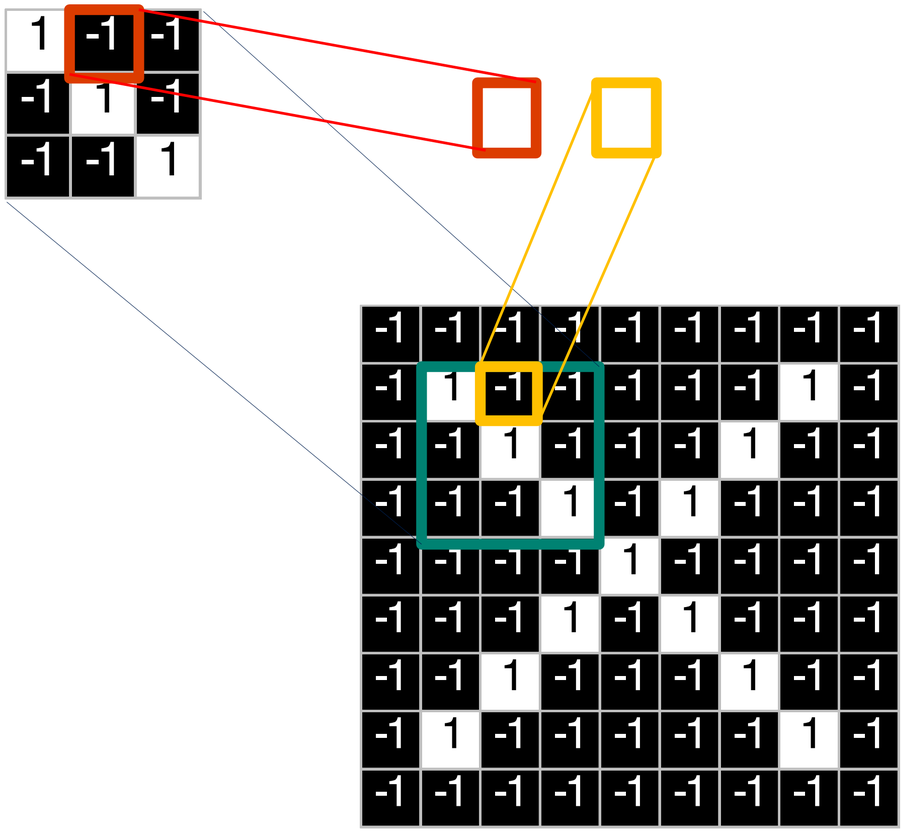
Filtering: The math behind the match

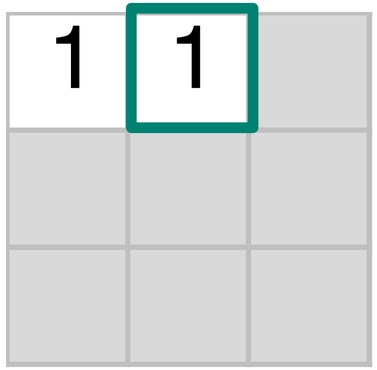




Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

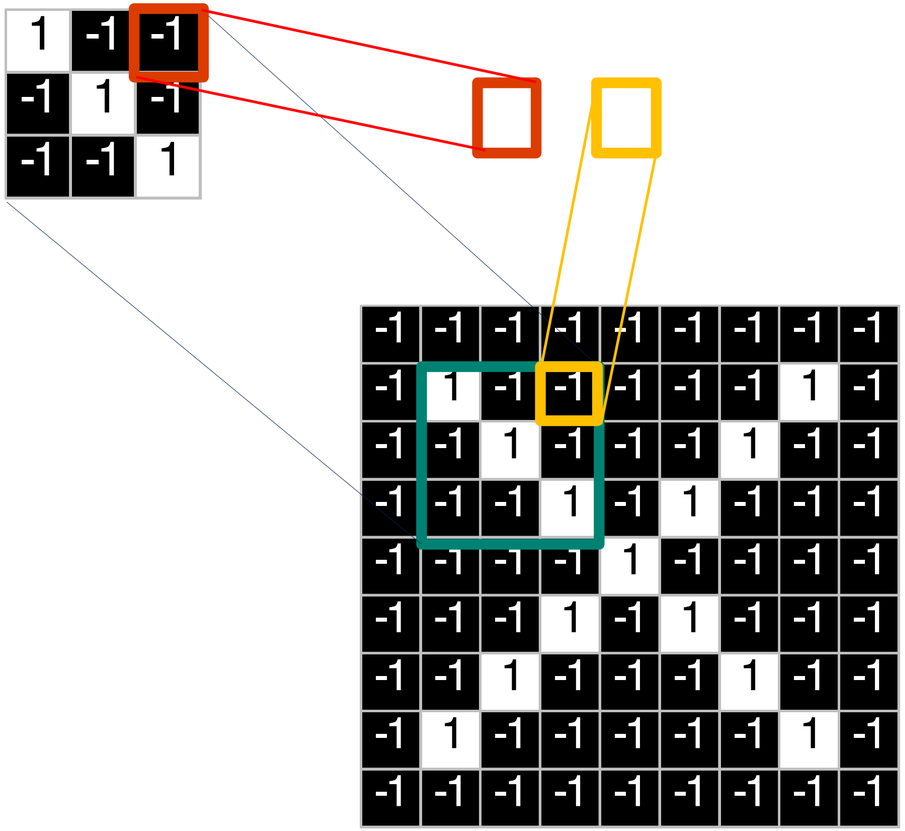
Filtering: The math behind the match

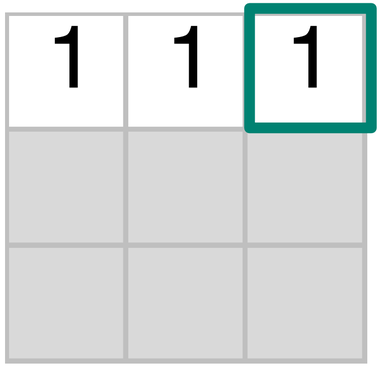




Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

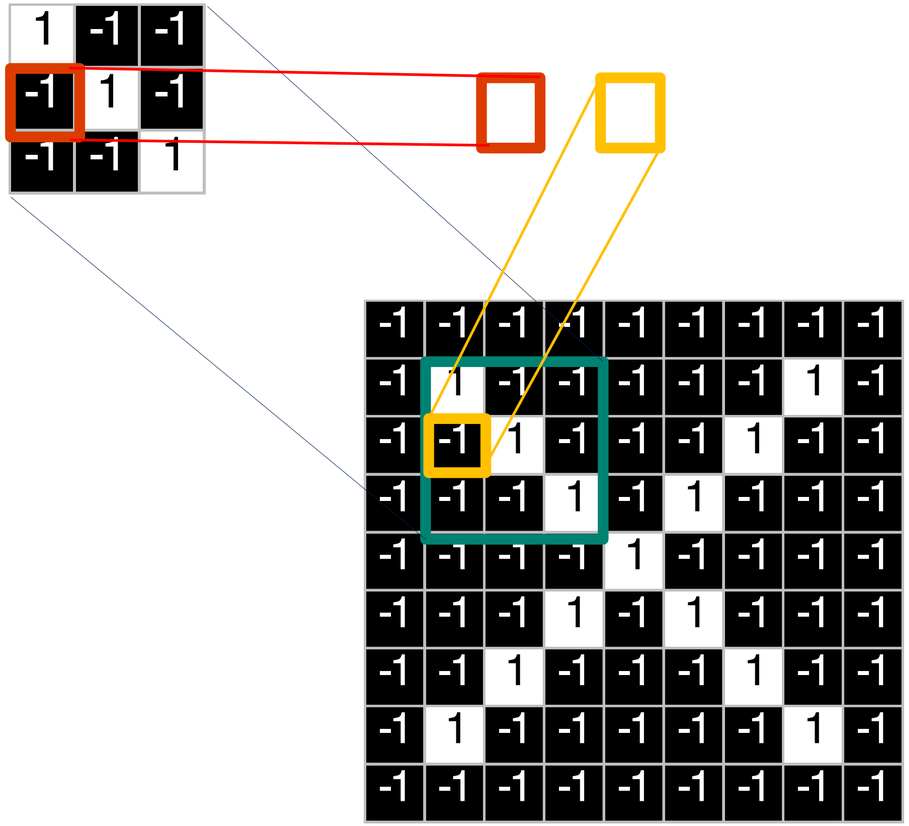
Filtering: The math behind the match

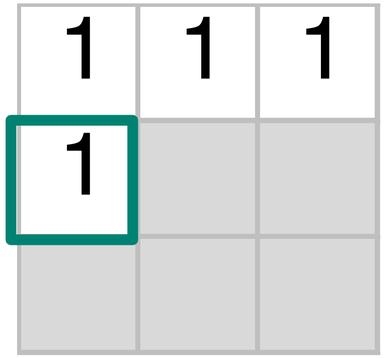




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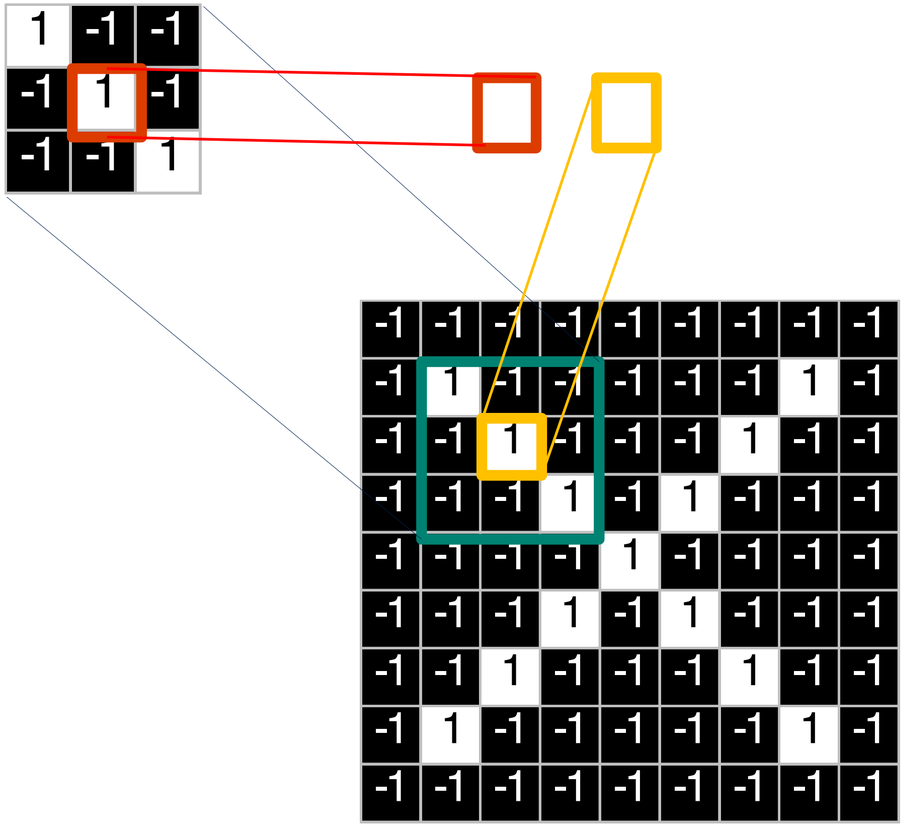
Filtering: The math behind the match

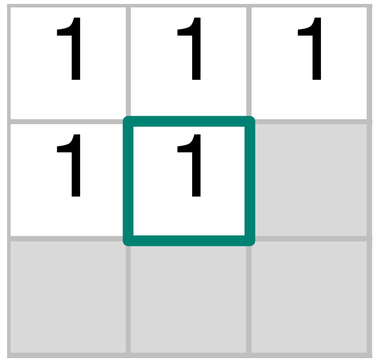




Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

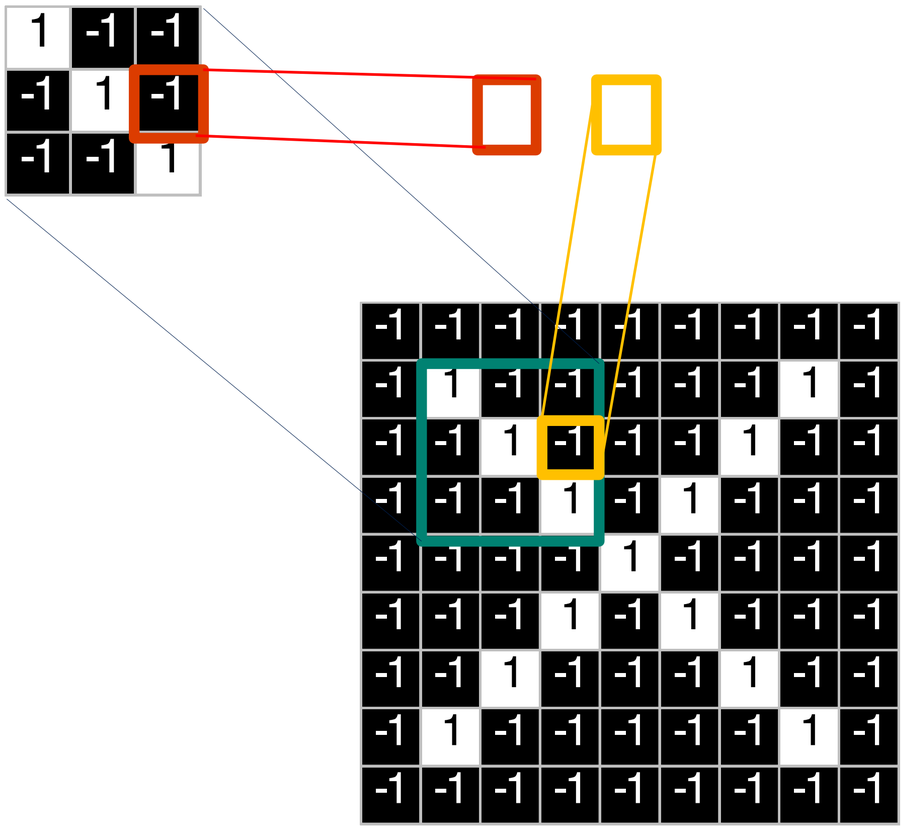
Filtering: The math behind the match

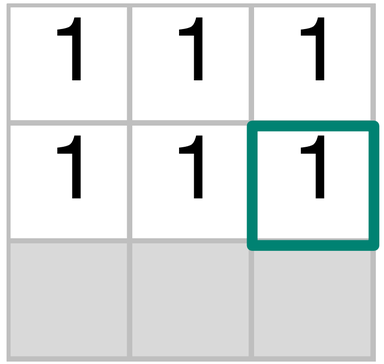




Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

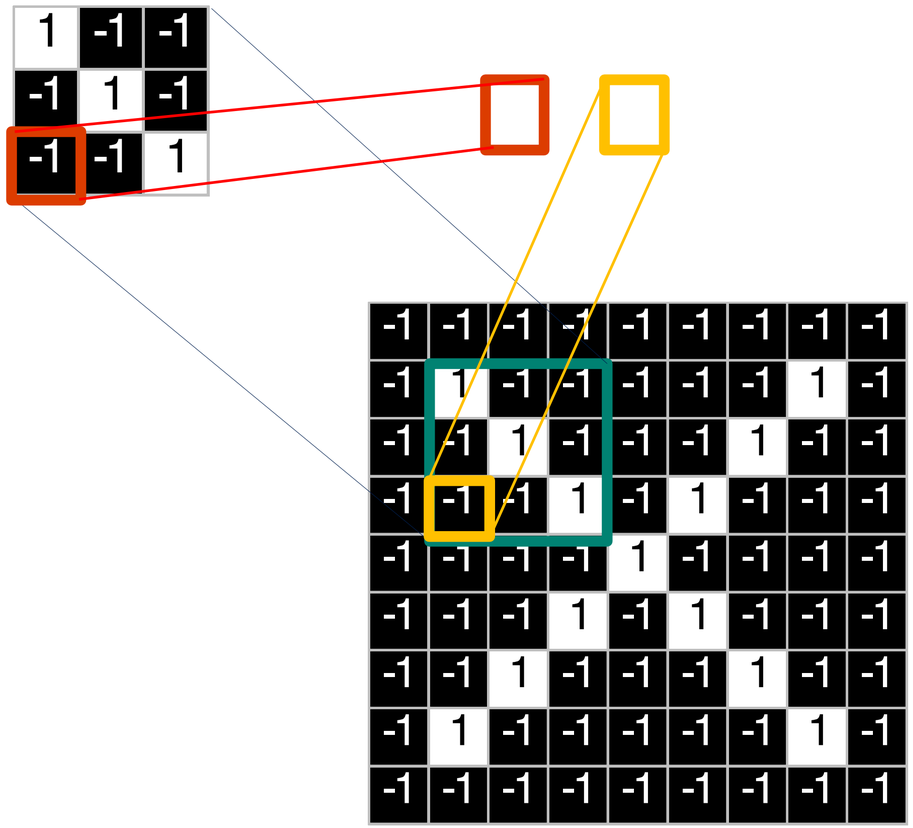
Filtering: The math behind the match





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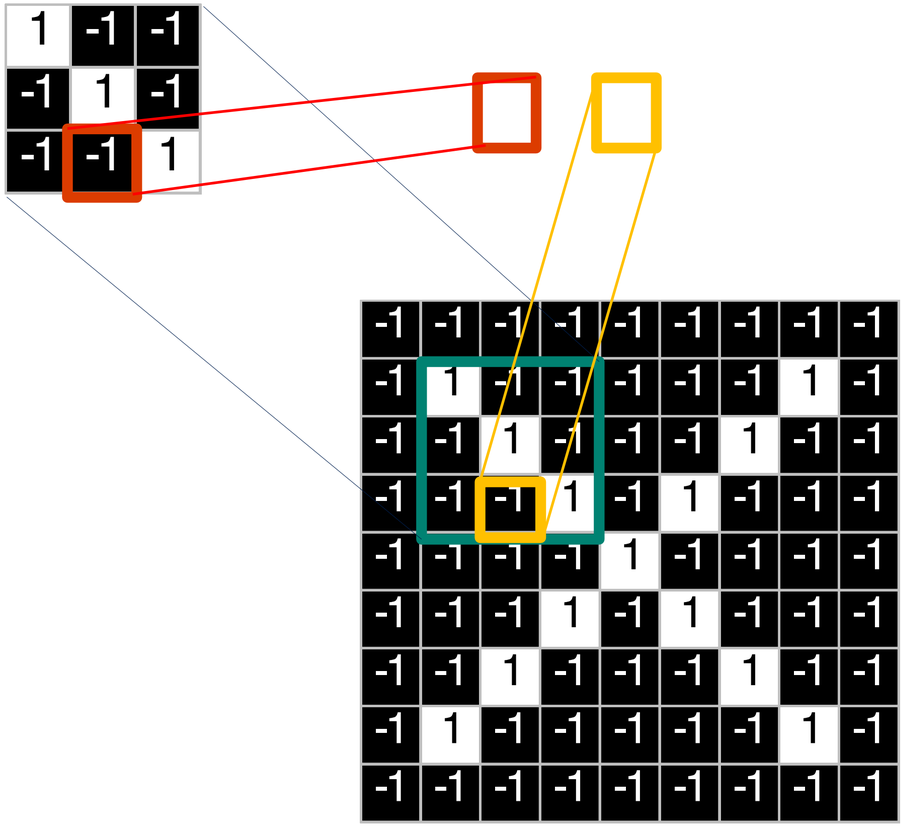
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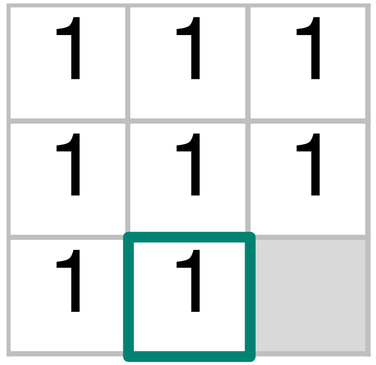


1	1	1
1	1	1
1		

Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

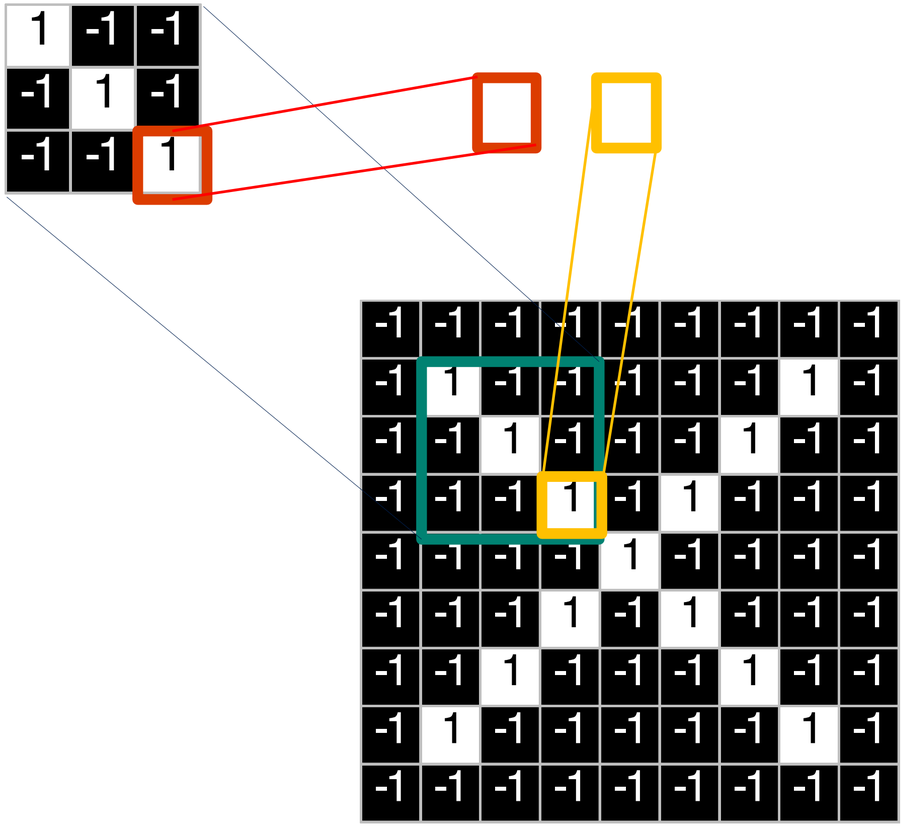
Filtering: The math behind the match

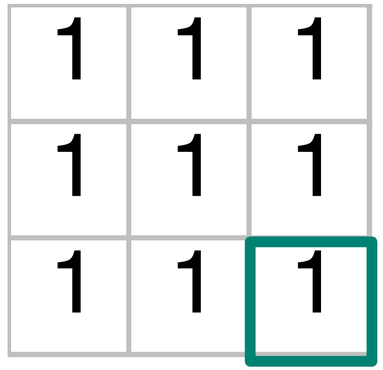




Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

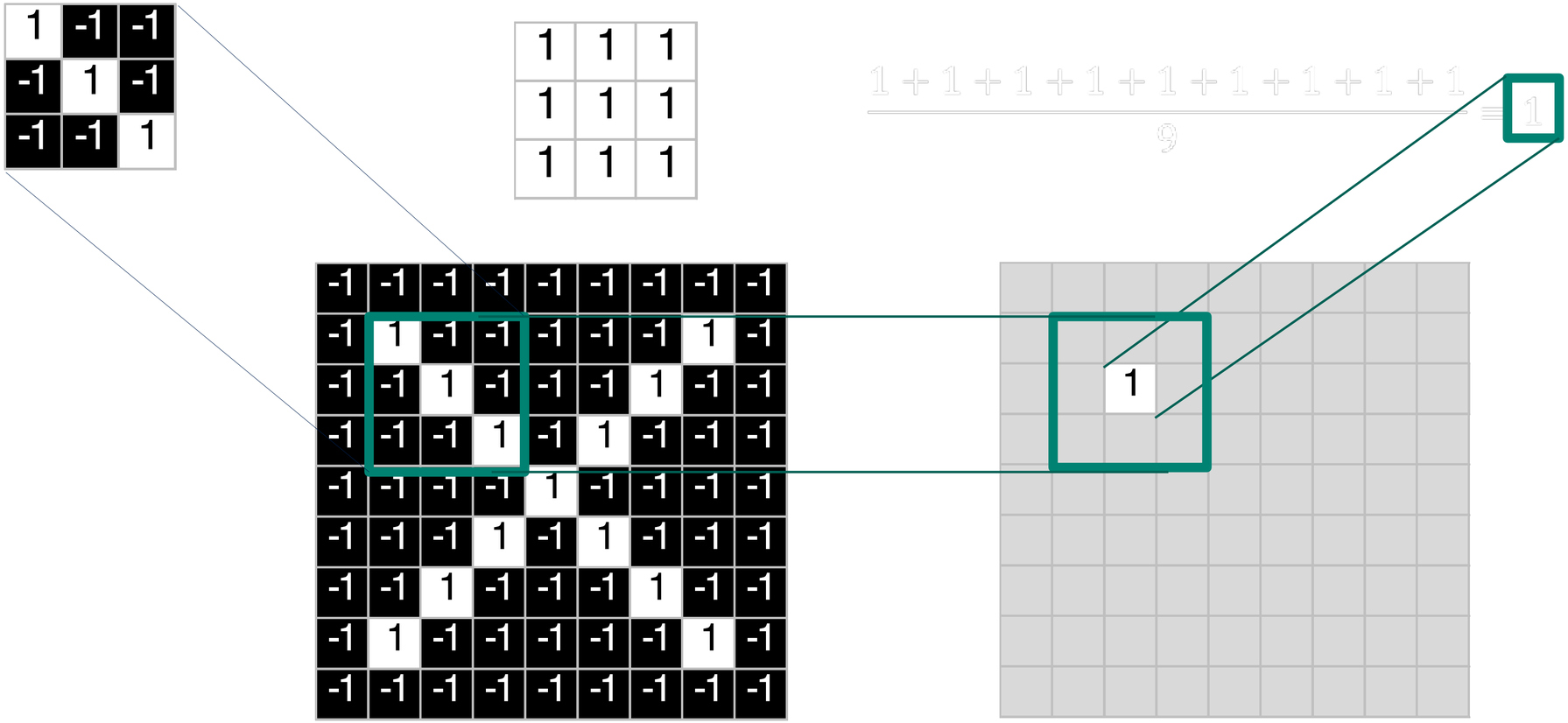
Filtering: The math behind the match





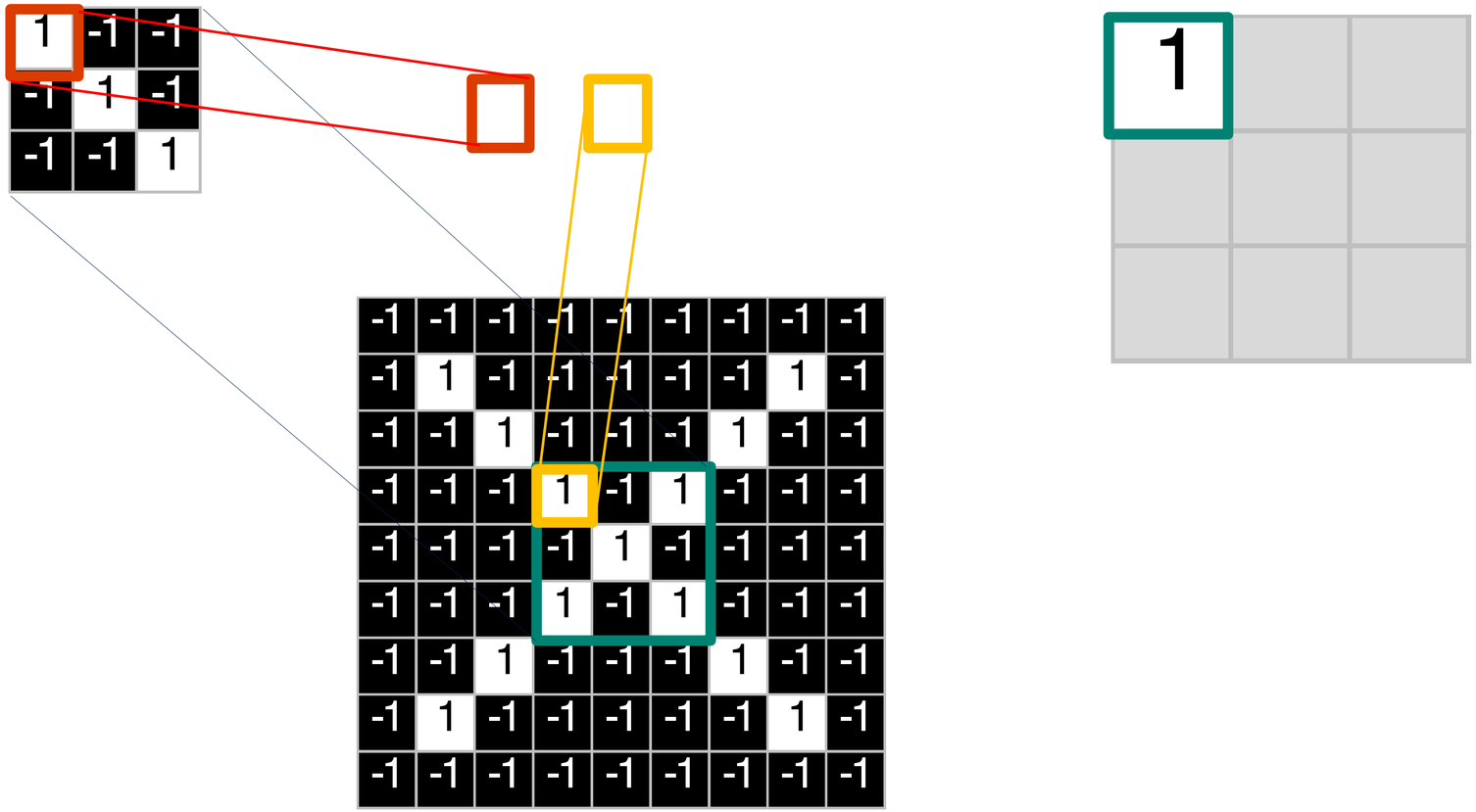
Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Filtering: The math behind the match



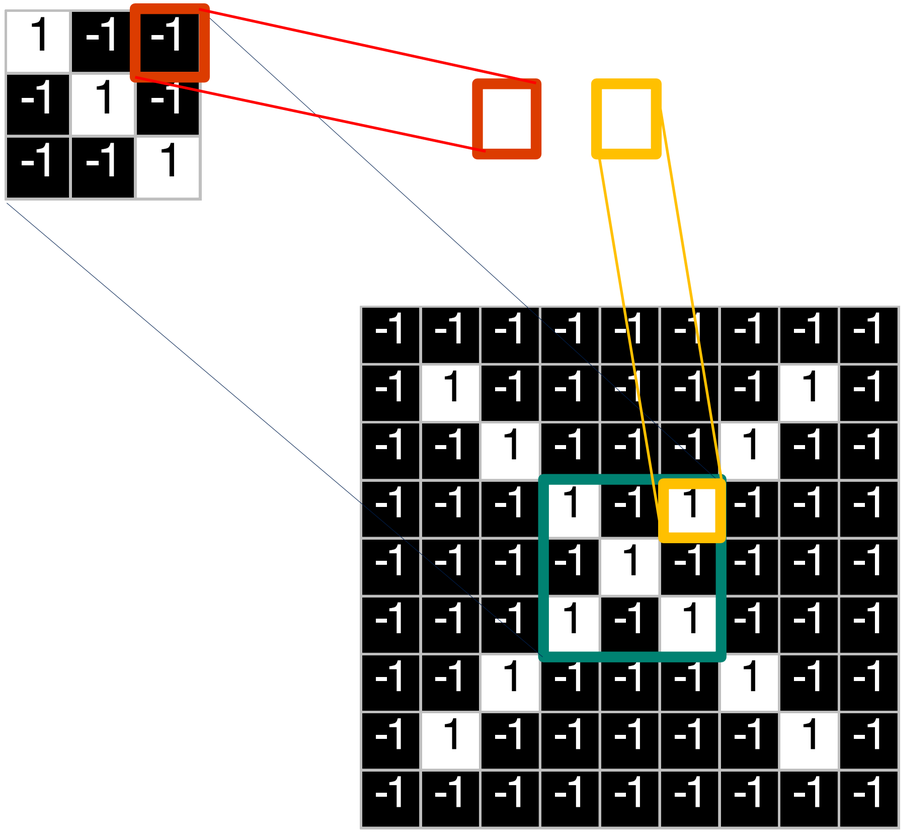
Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

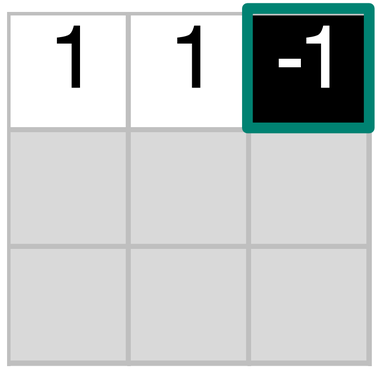
Filtering: The math behind the match



Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

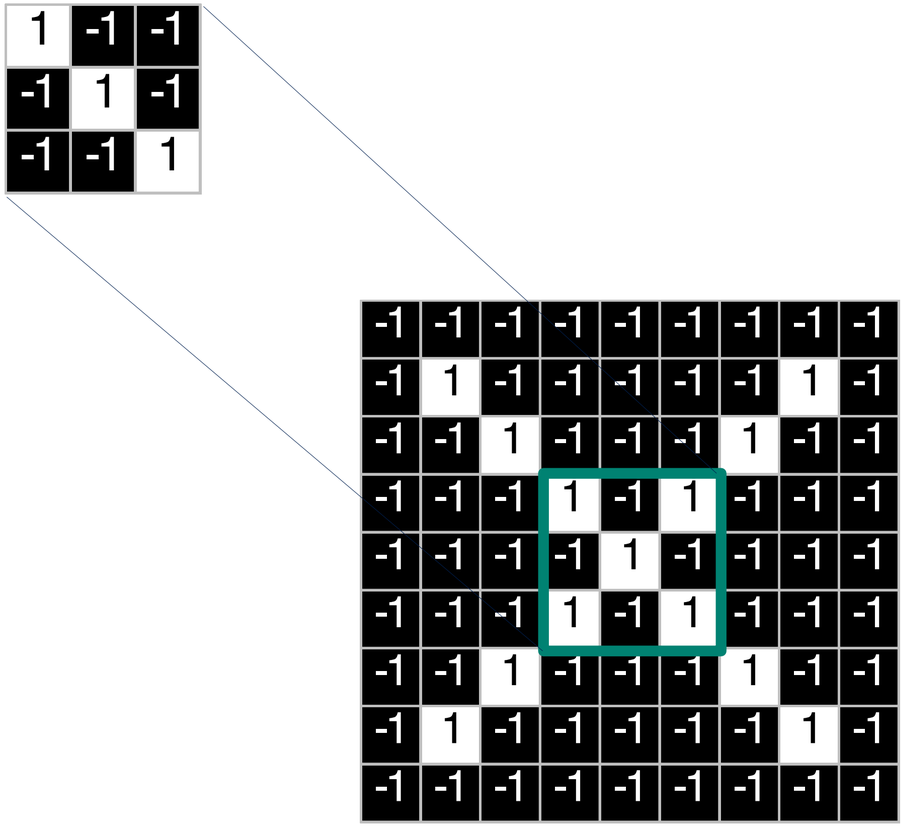
Filtering: The math behind the match





Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

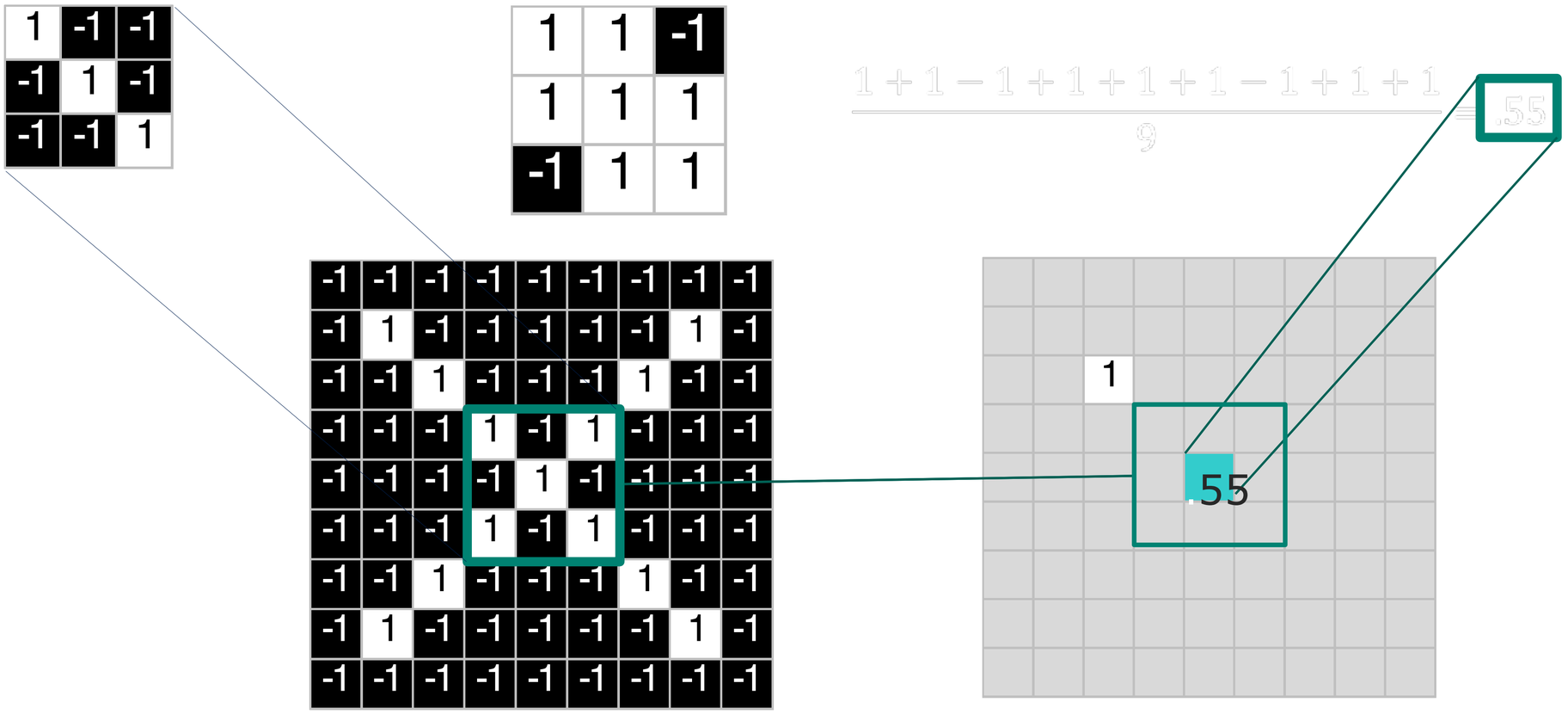
Filtering: The math behind the match



1	1	-1
1	1	1
-1	1	1

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Filtering: The math behind the match



Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Convolution: Trying every possible match

1	-1	-1
-1	1	-1
-1	-1	1

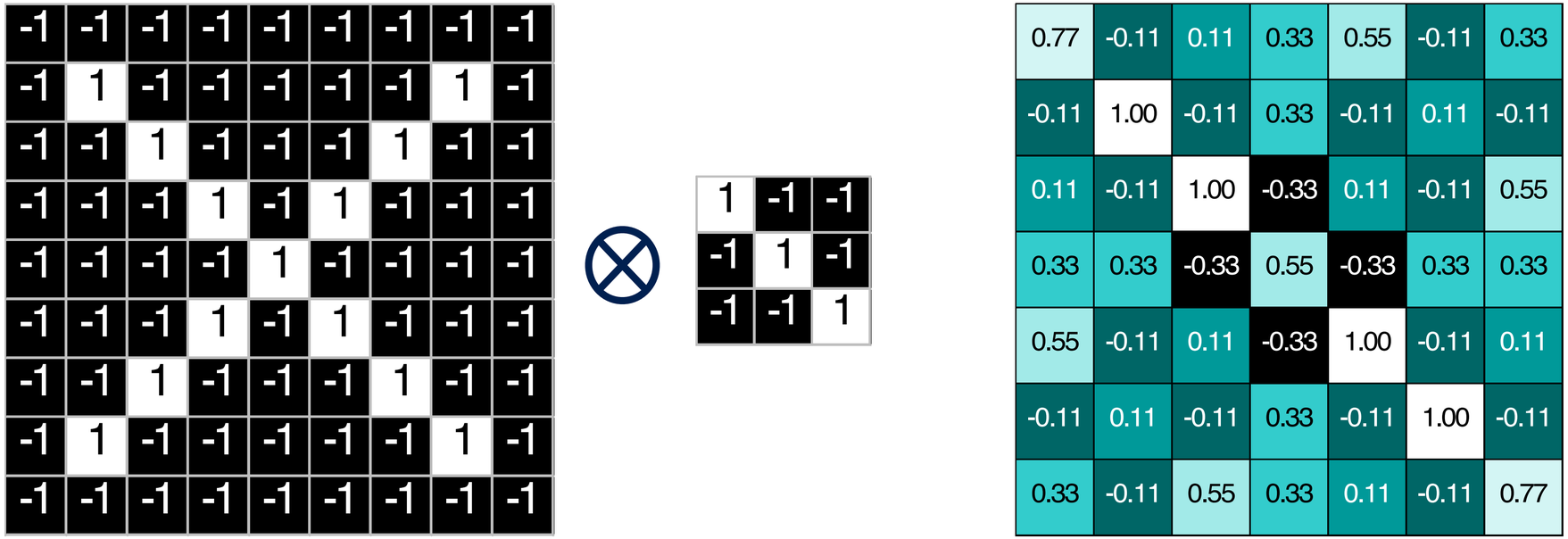
-1	-1	-1	-1	-1	-1	-1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1



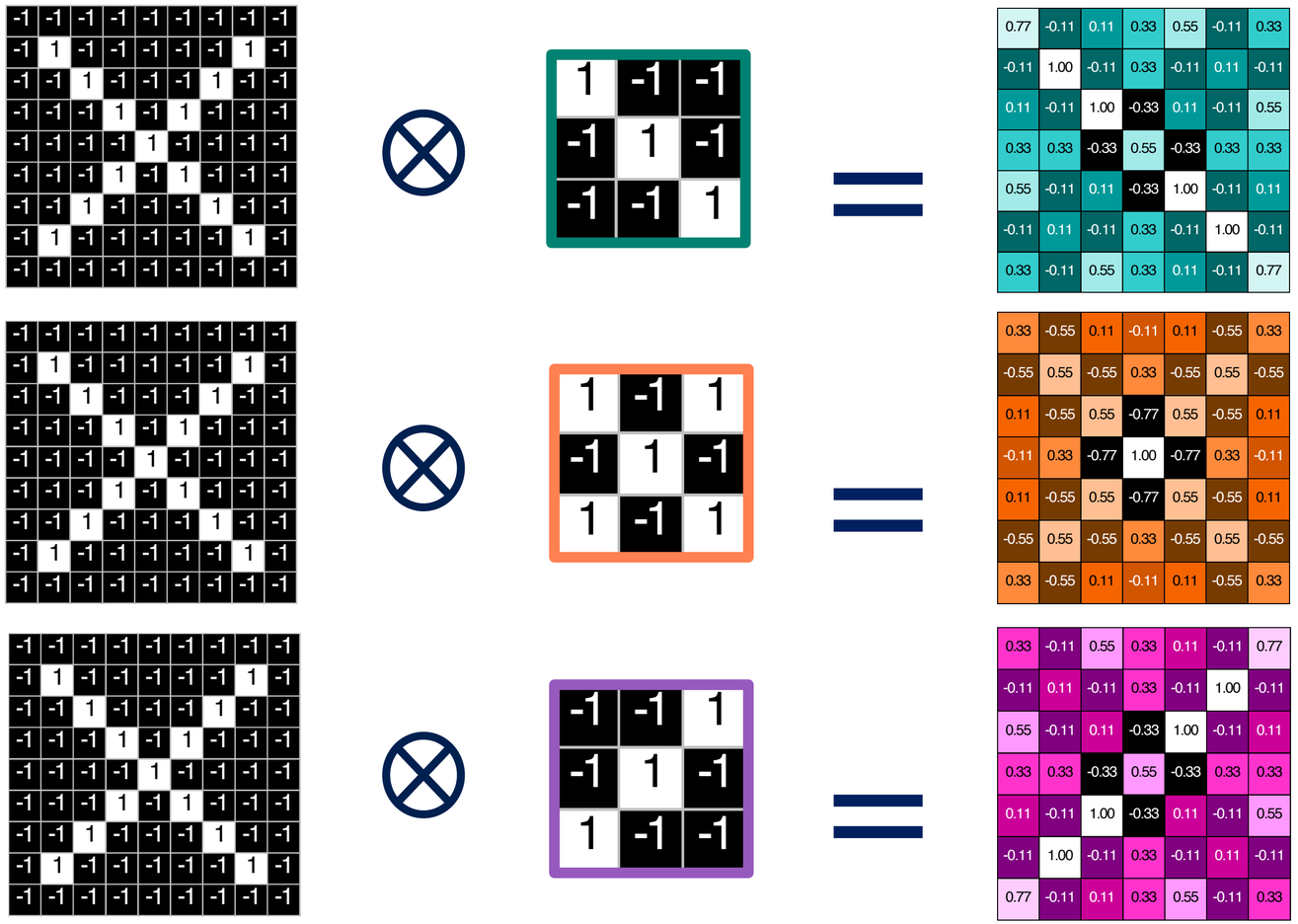
0.77	-0.11	0.11	0.33	0.55	-0.11	0.33
-0.11	1.00	-0.11	0.33	-0.11	0.11	-0.11
0.11	-0.11	1.00	-0.33	0.11	-0.11	0.55
0.33	0.33	-0.33	0.55	-0.33	0.33	0.33
0.55	-0.11	0.11	-0.33	1.00	-0.11	0.11
-0.11	0.11	-0.11	0.33	-0.11	1.00	-0.11
0.33	-0.11	0.55	0.33	0.11	-0.11	0.77

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Convolution: Trying every possible match



Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

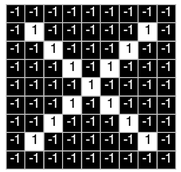


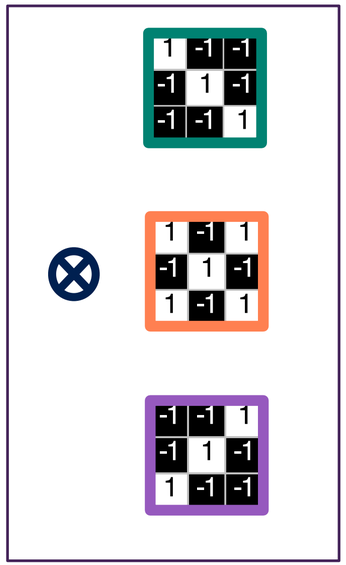
Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

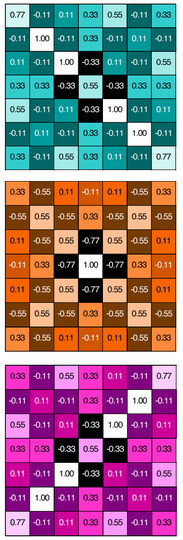
Convolution layer

Foundations of AI AI500

One image becomes a stack of filtered images



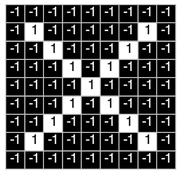


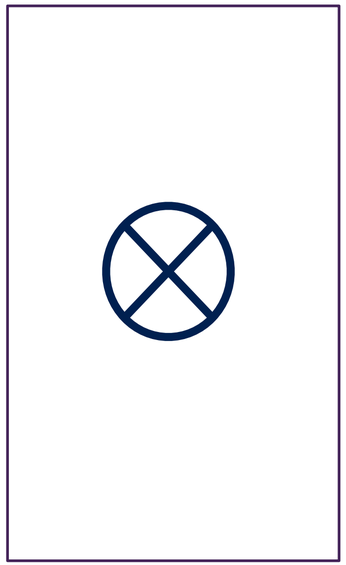


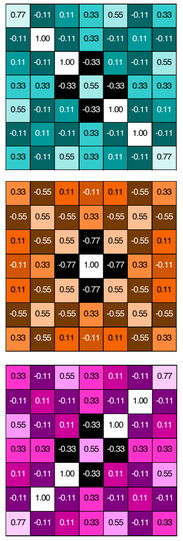
Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Convolution layer

One image becomes a stack of filtered images







Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

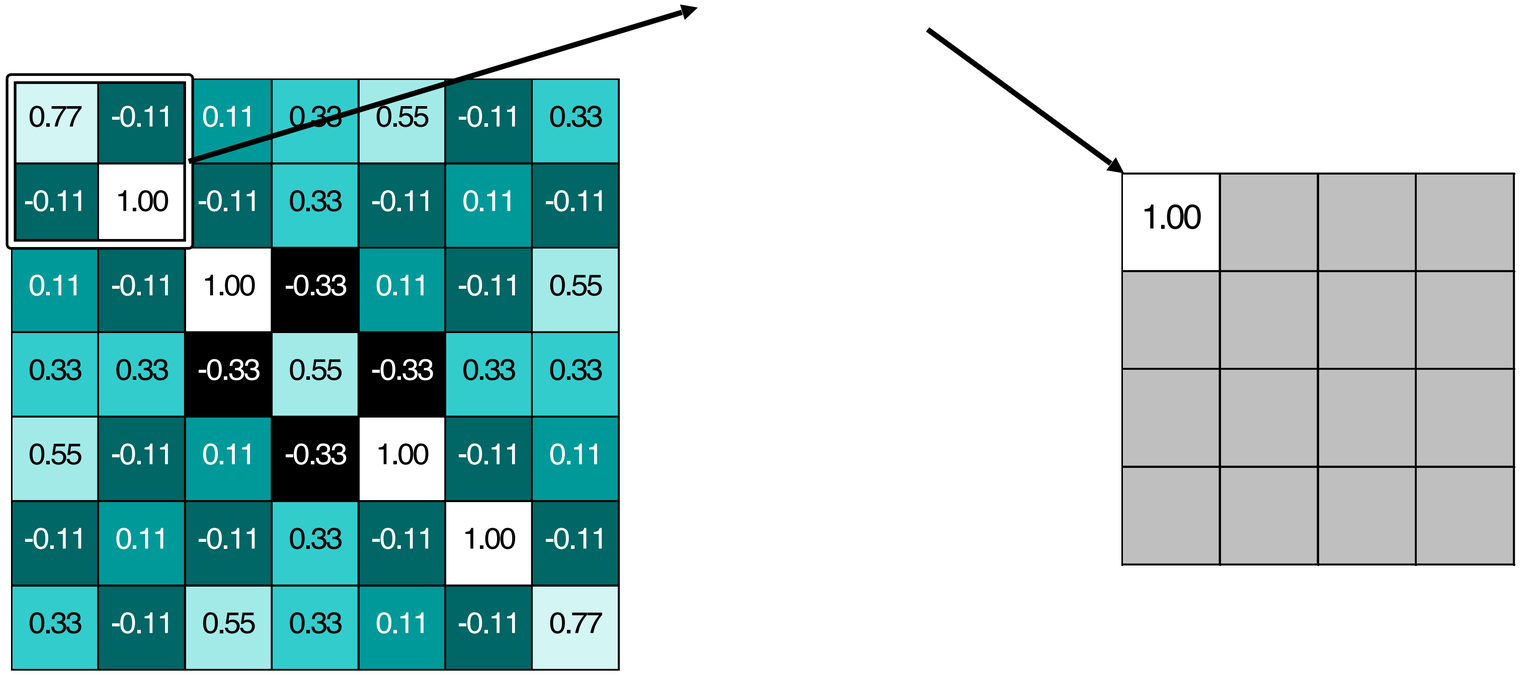
Pooling: Shrinking the image stack



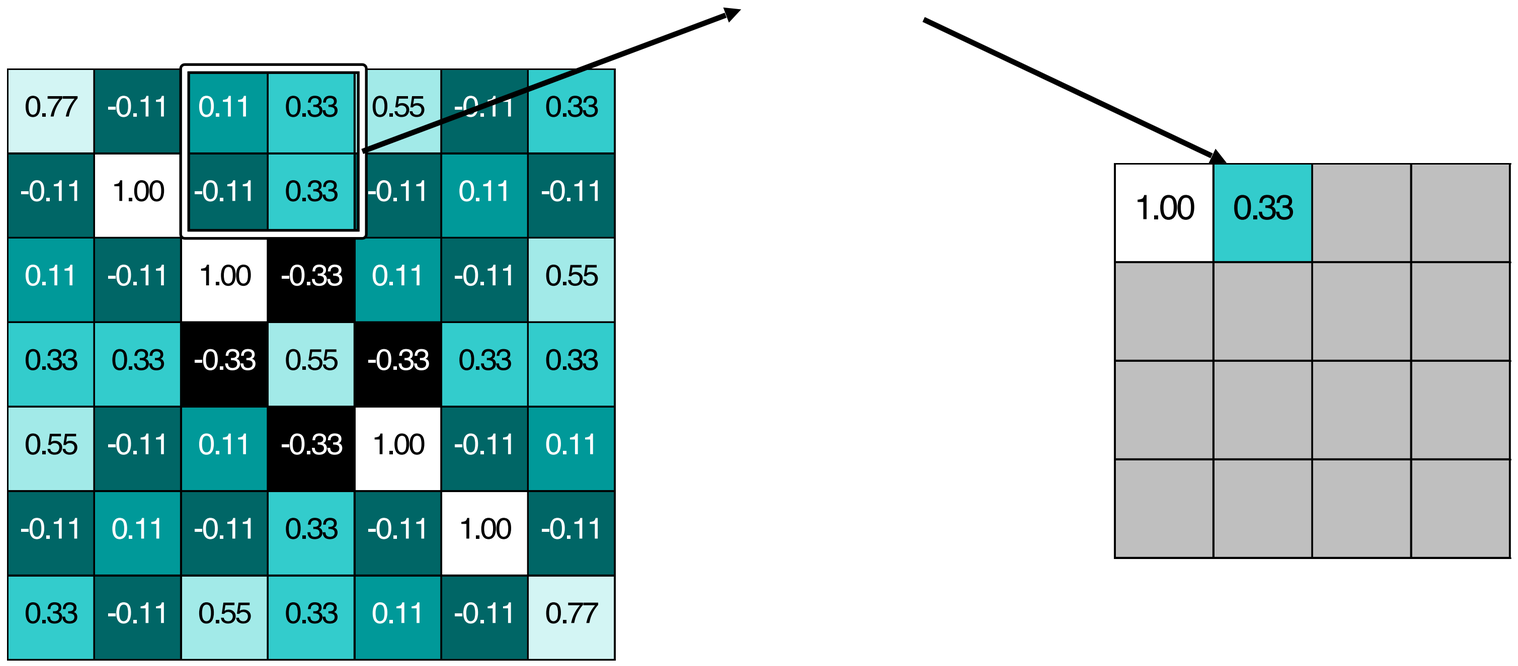
1. Pick a window size (usually 2 or 3).
2. Pick a stride (usually 2).
3. Walk your window across your filtered images.
4. From each window, take the maximum value.

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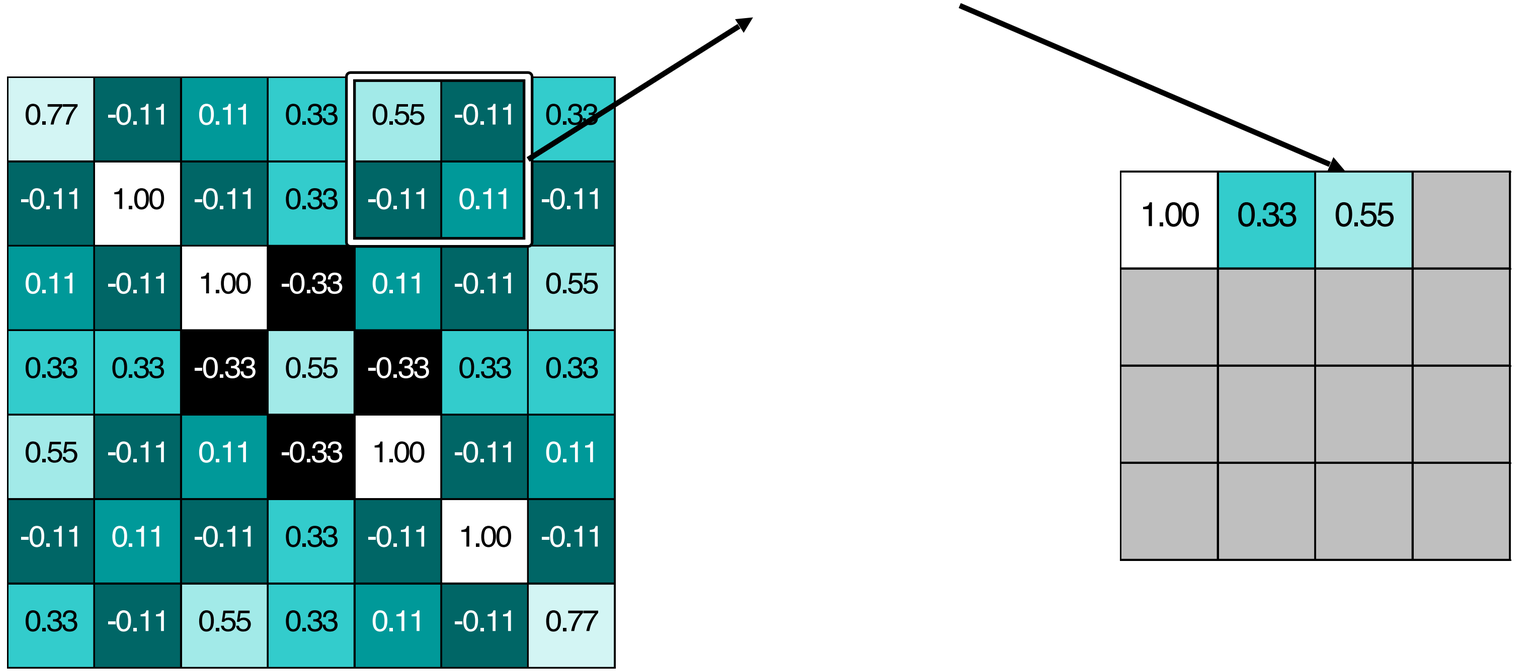
Pooling


0.77 -0.11 0.11 0.33 0.55 -0.11 0.33
-0.11 1.00 -0.11 0.33 -0.11 0.11 -0.11
0.11 -0.11 1.00 -0.33 0.11 -0.11 0.55
0.33 0.33 -0.33 0.55 -0.33 0.33 0.33
0.55 -0.11 0.11 -0.33 1.00 -0.11 0.11
-0.11 0.11 -0.11 0.33 -0.11 1.00 -0.11
0.33 -0.11 0.55 0.33 0.11 -0.11 0.77
1.00

Pooling

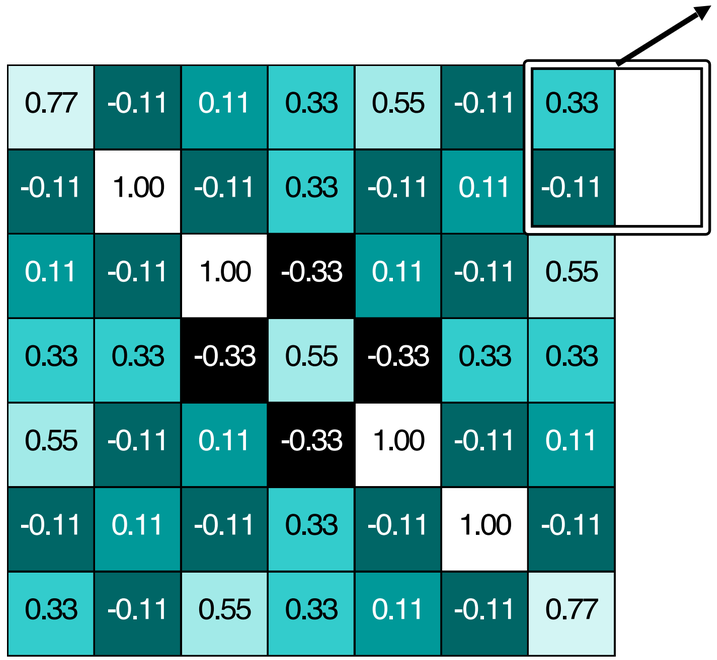

0.77 -0.11 0.11 0.33 0.55 -0.11 0.33
-0.11 1.00 -0.11 0.33 -0.11 0.11 -0.11
0.11 -0.11 1.00 -0.33 0.11 -0.11 0.55
0.33 0.33 -0.33 0.55 -0.33 0.33 0.33
0.55 -0.11 0.11 -0.33 1.00 -0.11 0.11
-0.11 0.11 -0.11 0.33 -0.11 1.00 -0.11
0.33 -0.11 0.55 0.33 0.11 -0.11 0.77
1.00 0.33

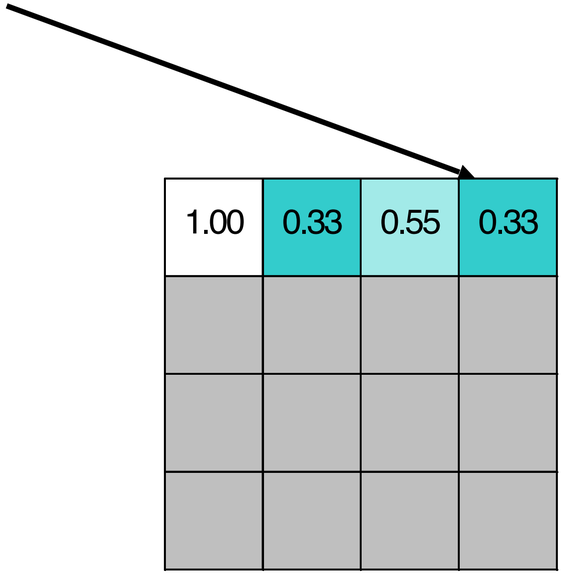
Pooling



Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

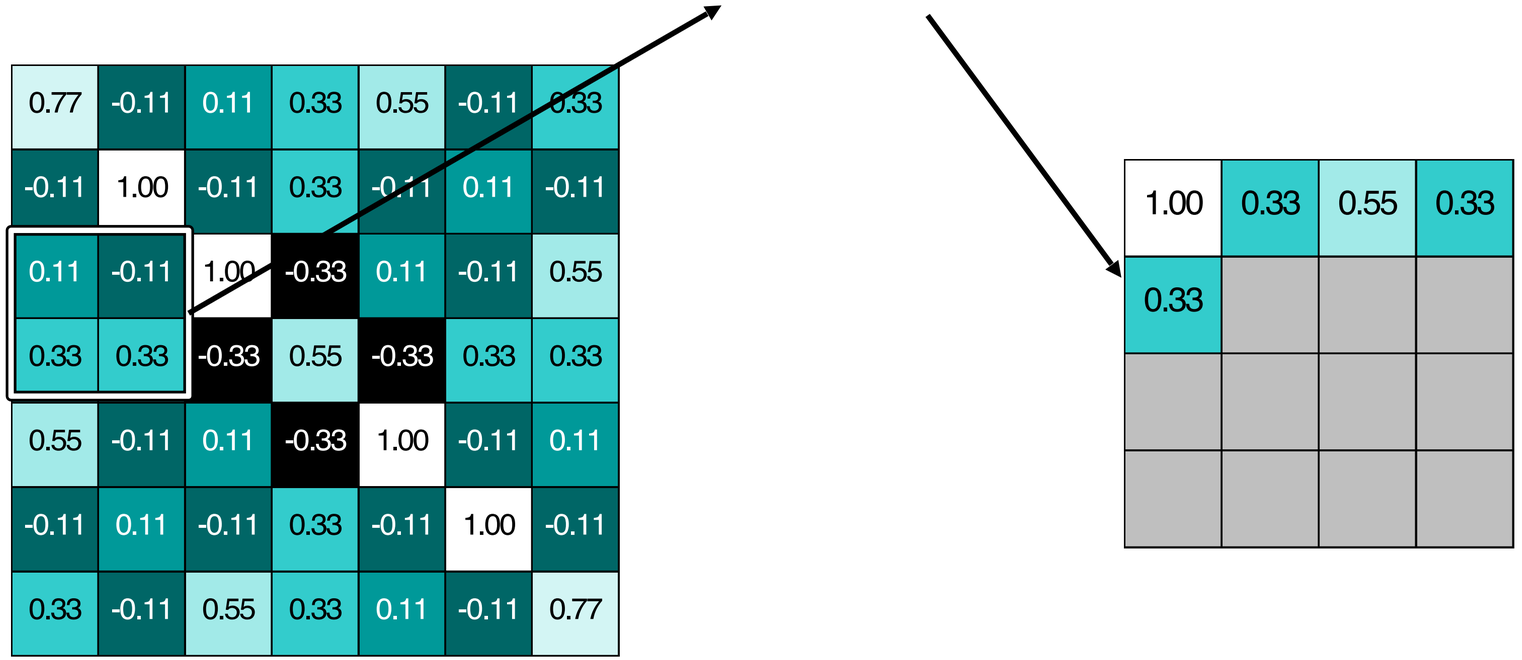
Pooling





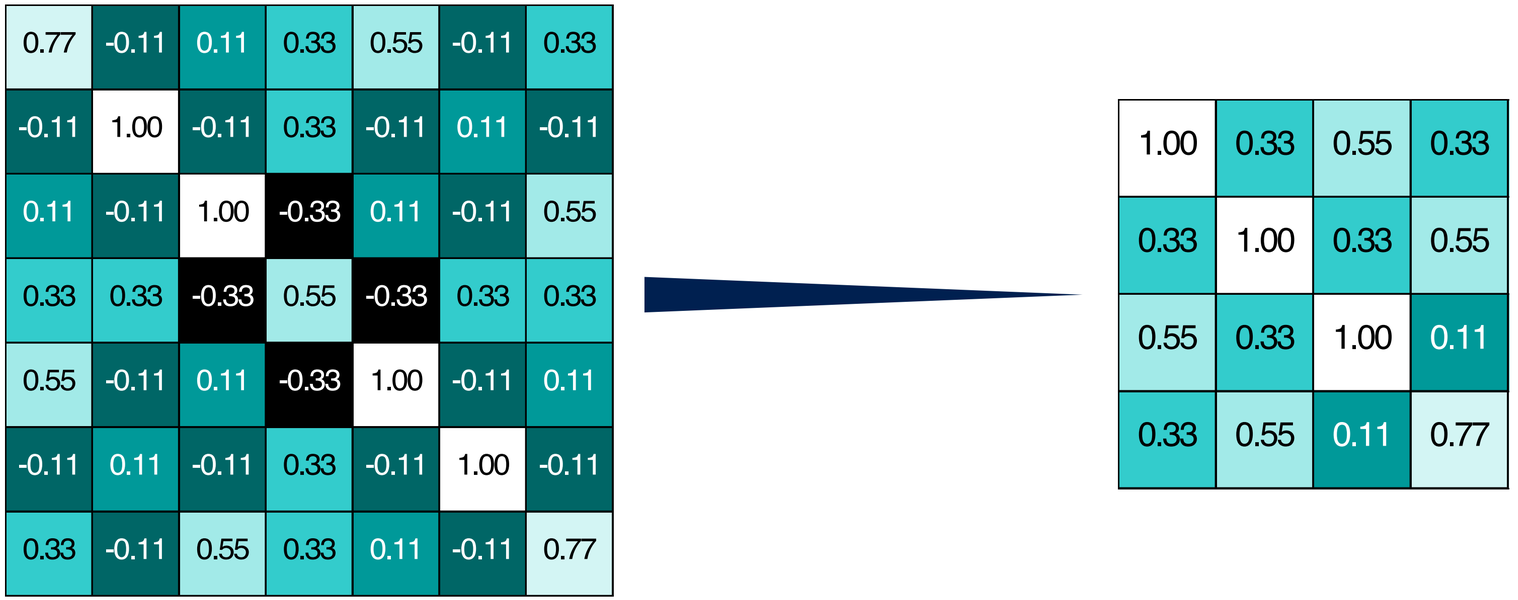
Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Pooling

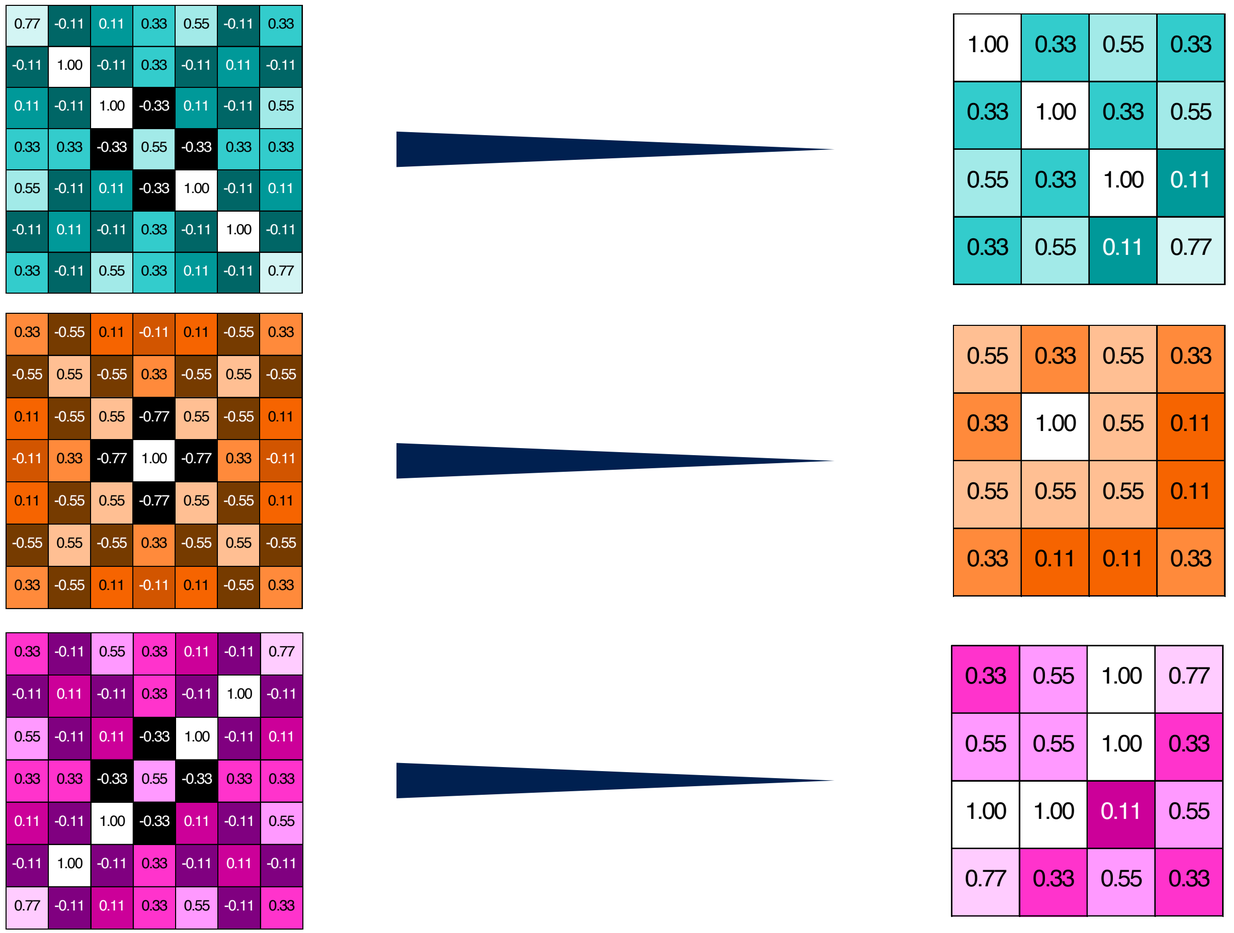


Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Pooling

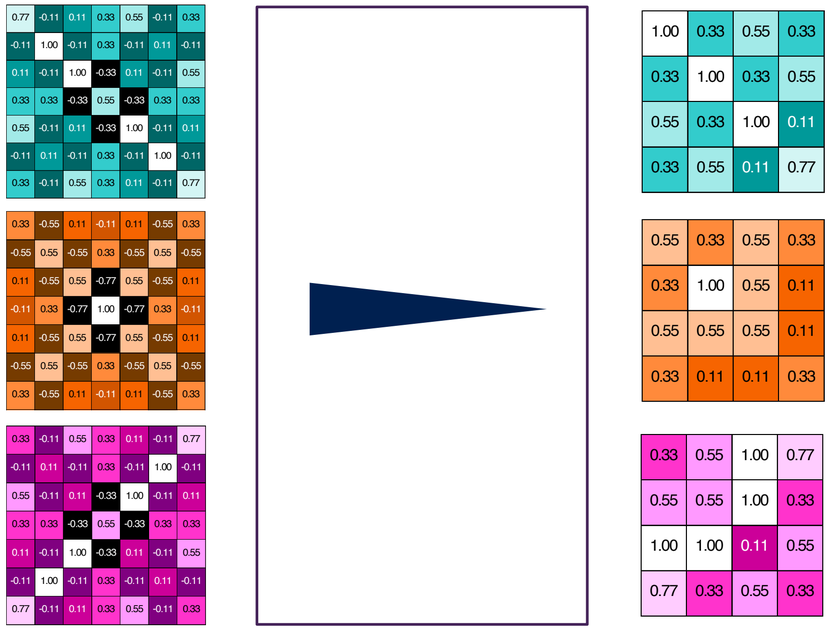


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Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Pooling layer



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Normalization

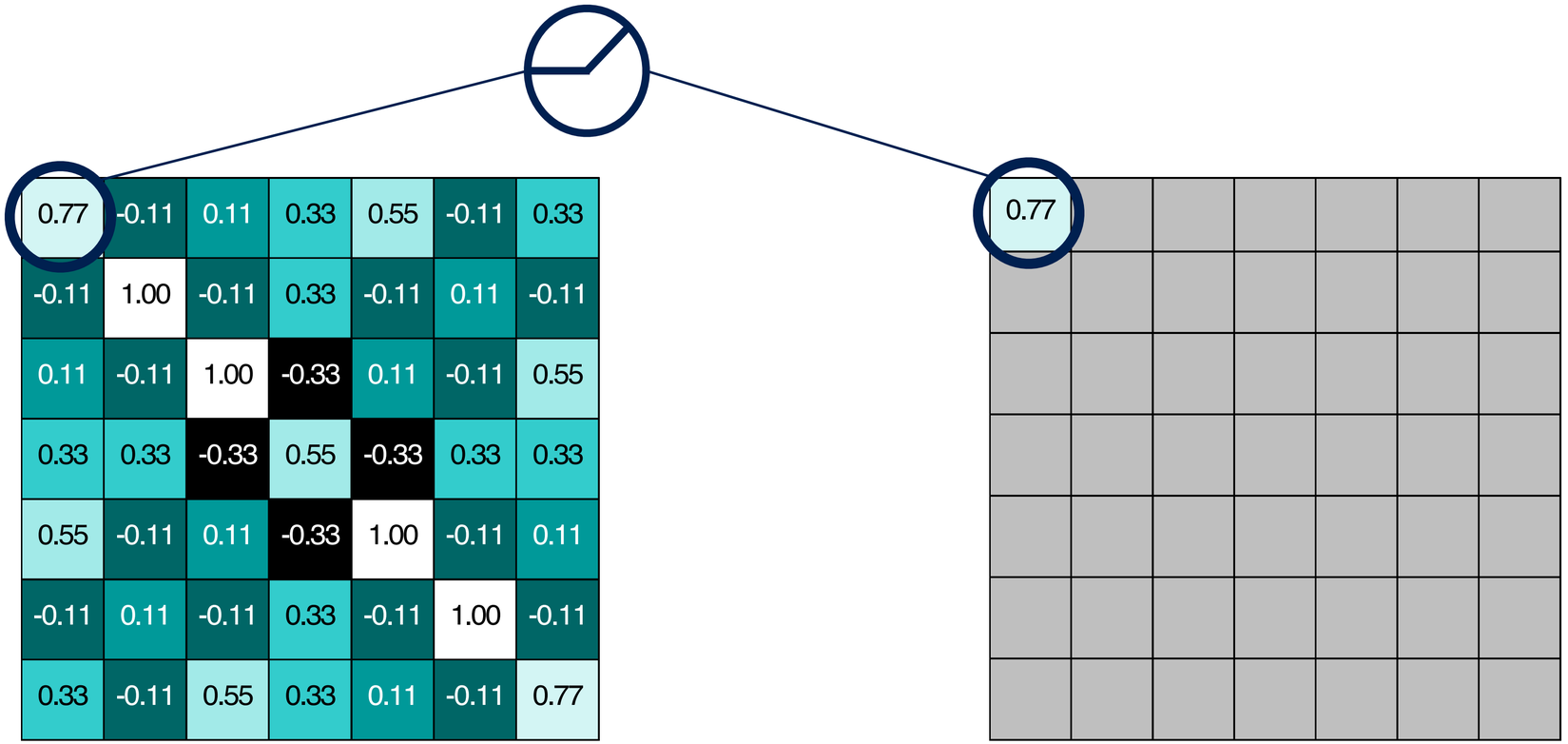


Keep the math from breaking by tweaking each of the values just a bit.

Change everything negative to zero.

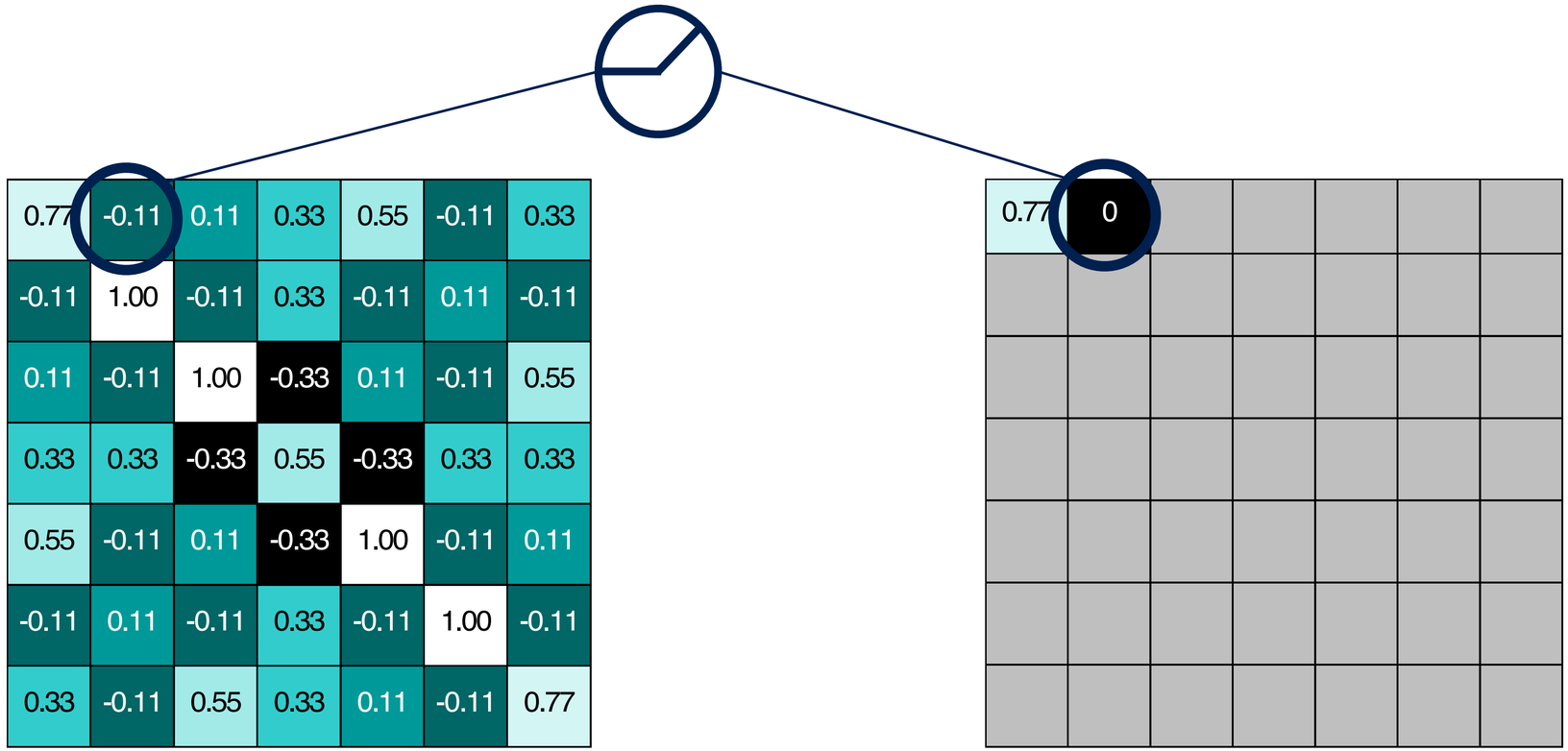
Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Rectified Linear Units (ReLUs)



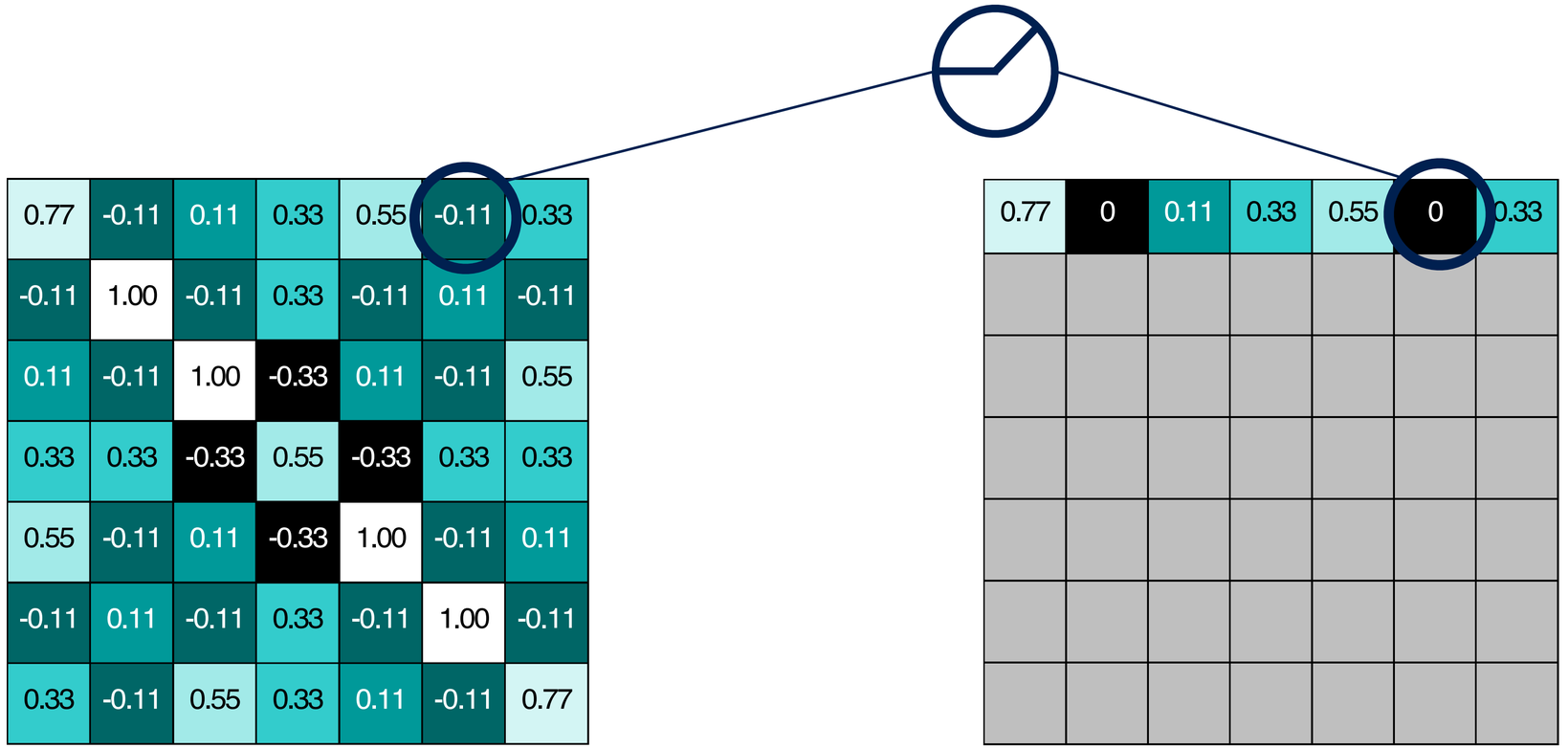
Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Rectified Linear Units (ReLUs)



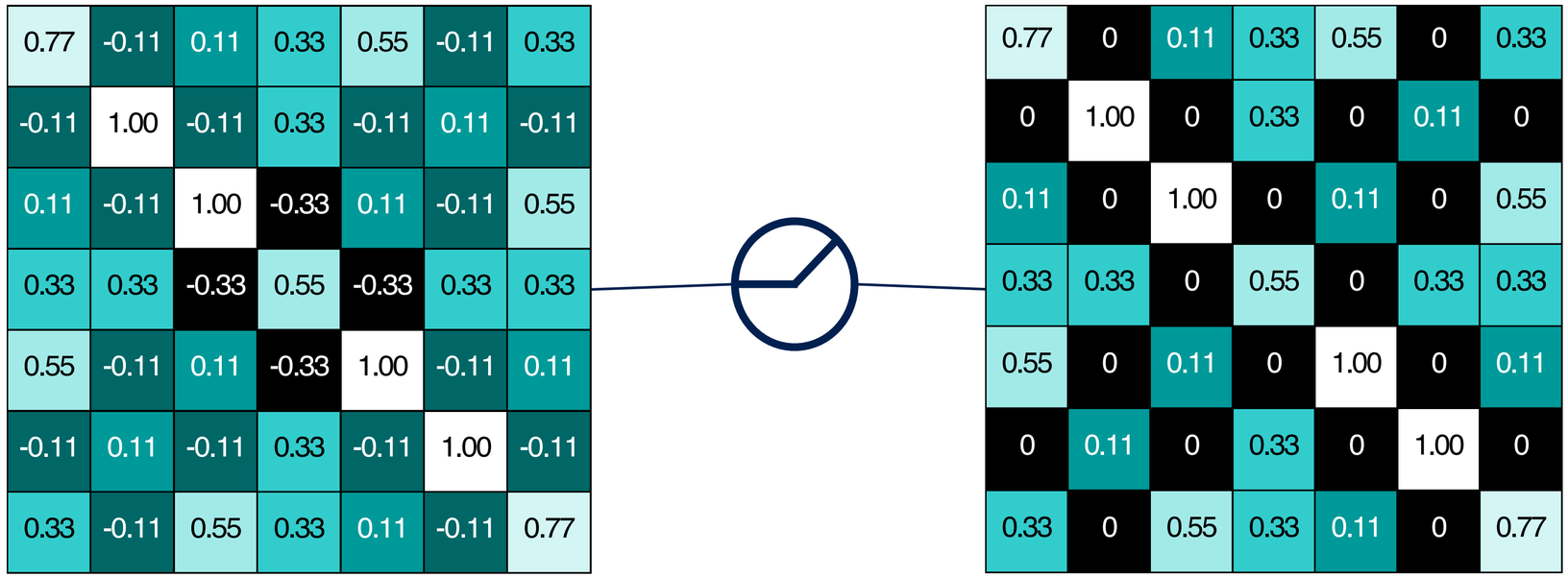
Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Rectified Linear Units (ReLUs)



Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

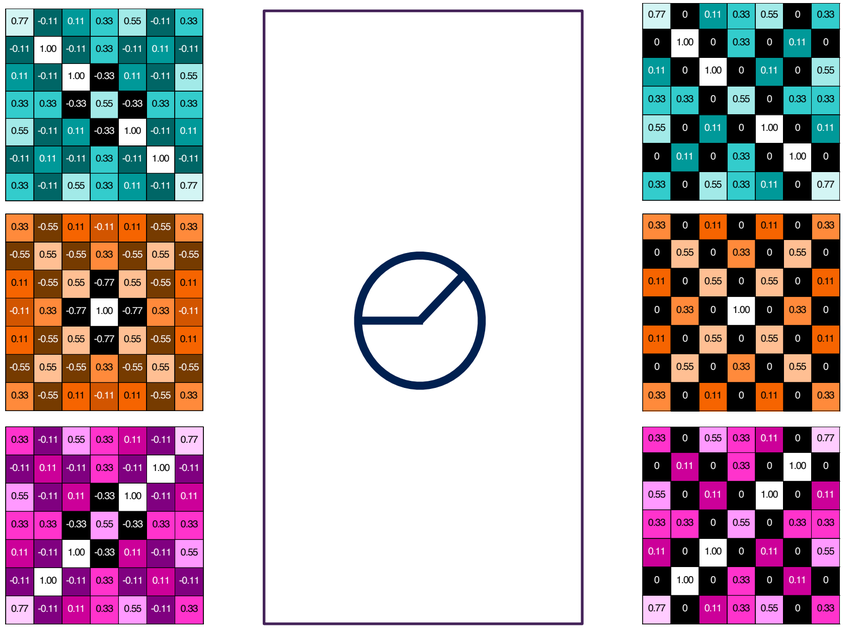
Rectified Linear Units (ReLUs)



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ReLU layer

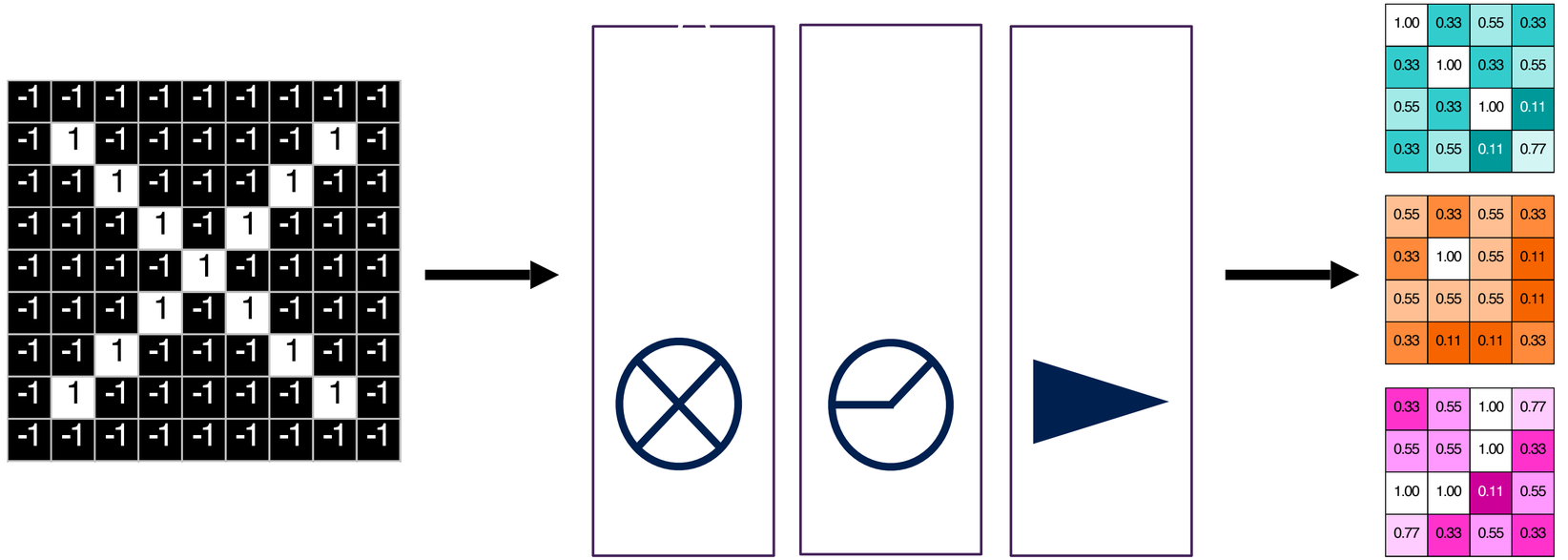
A stack of images becomes a stack of images with no negative values.



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Layers get stacked

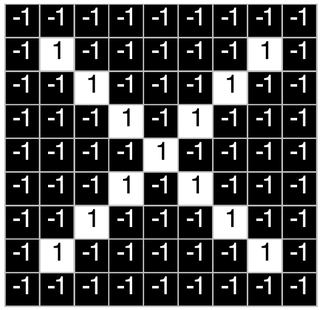
The output of one becomes the input of the next.

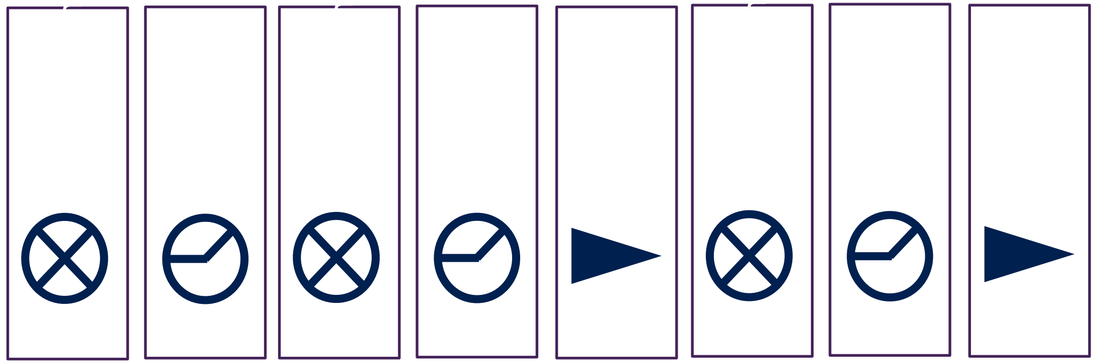


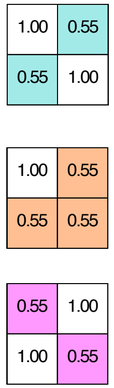
Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Deep stacking

Layers can be repeated several (or many) times.



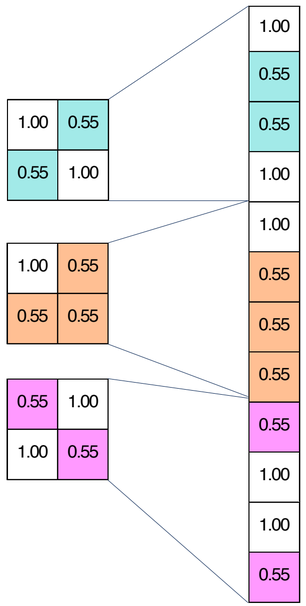




Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Fully connected layer

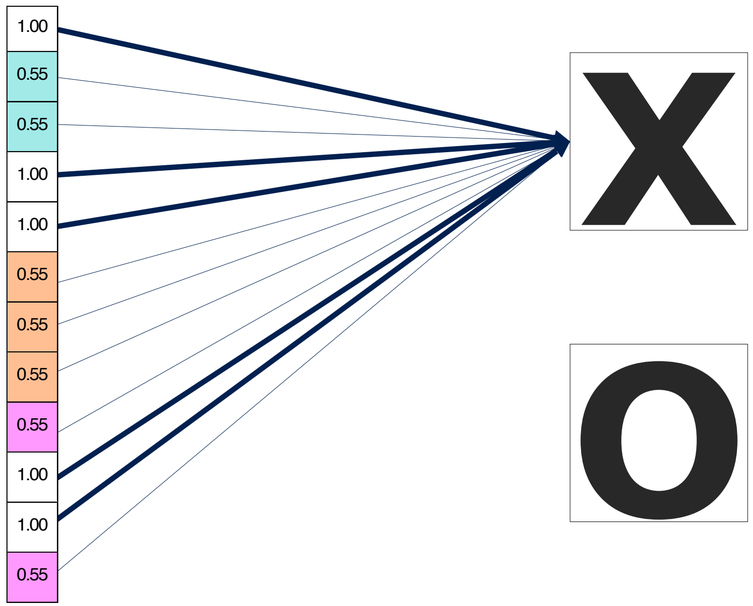
Every value gets a vote



Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Fully connected layer

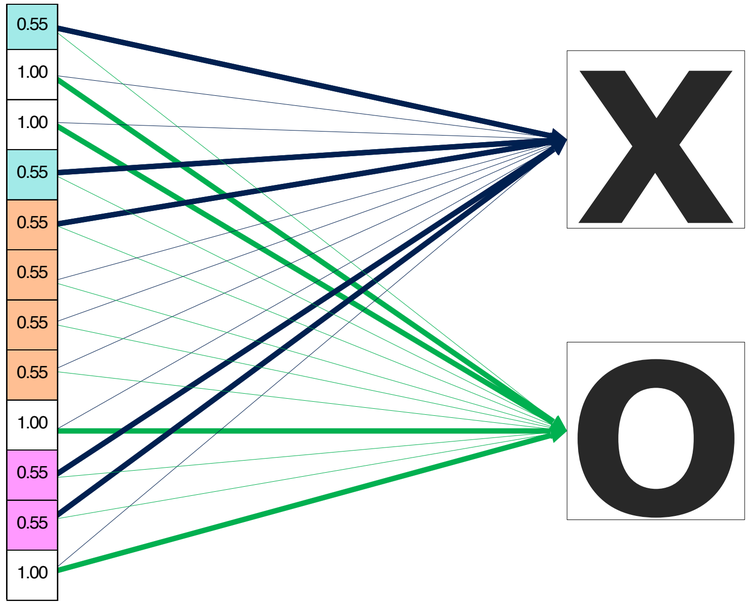
Vote depends on how strongly a value predicts X or O



Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Fully connected layer

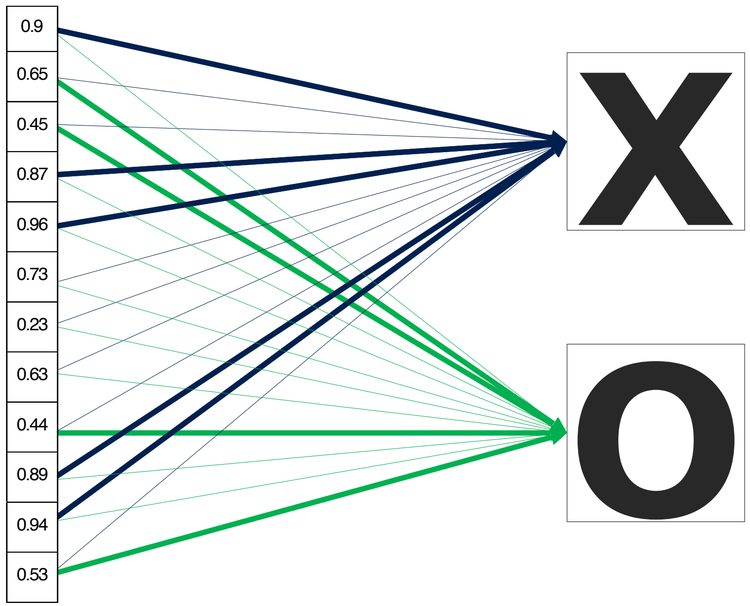
Vote depends on how strongly a value predicts X or O



Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Fully connected layer

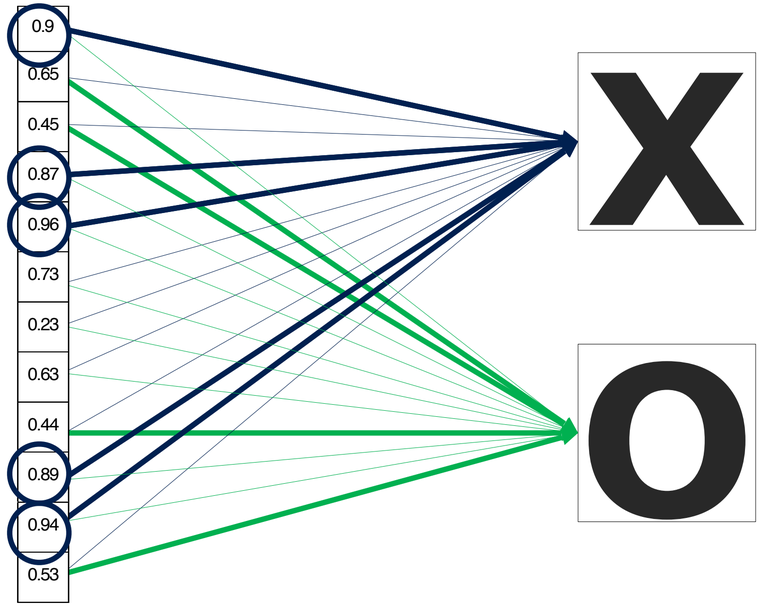
Future values vote on X or O



Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Fully connected layer

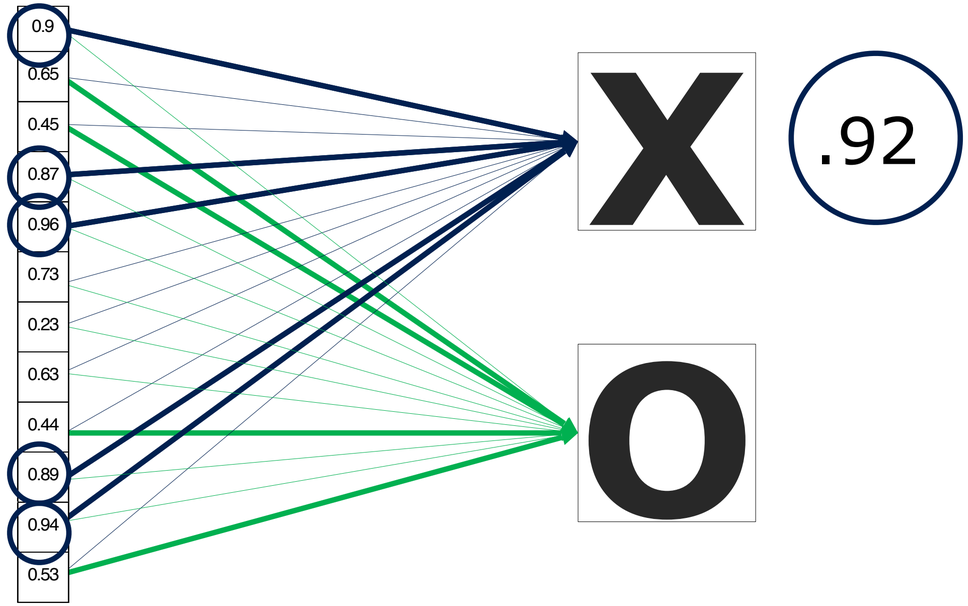
Future values vote on X or O



Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Fully connected layer

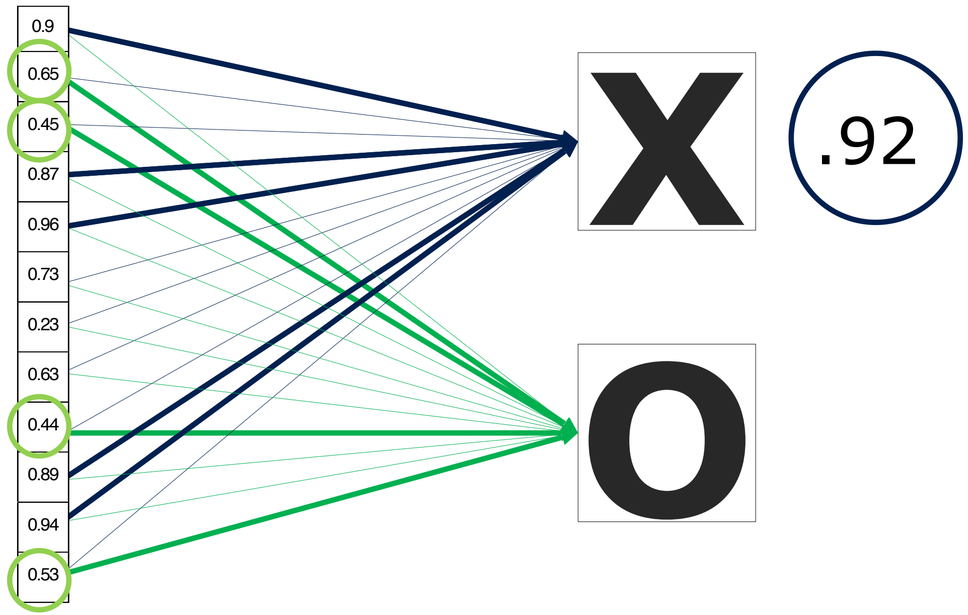
Future values vote on X or O



Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Fully connected layer

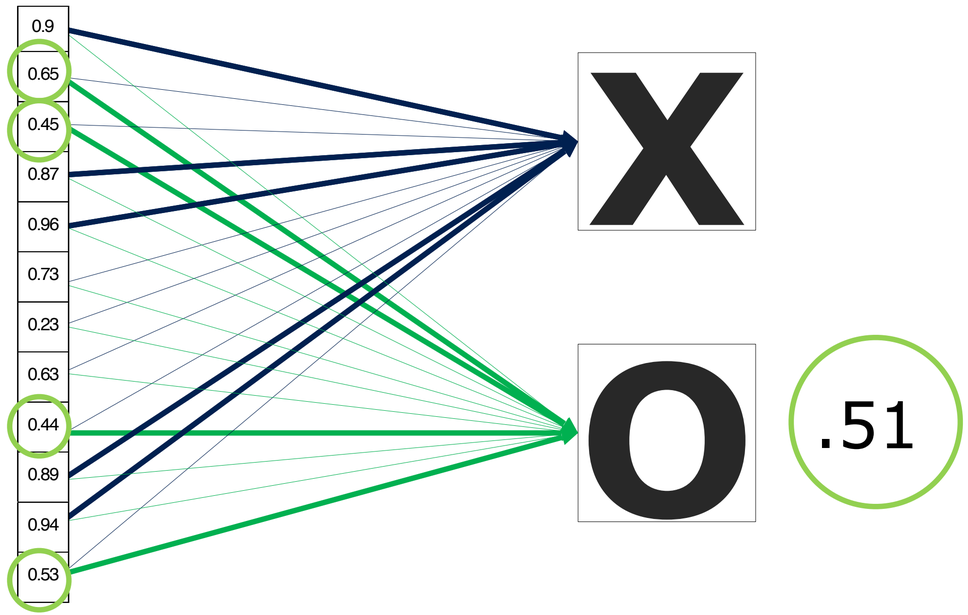
Future values vote on X or O



Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Fully connected layer

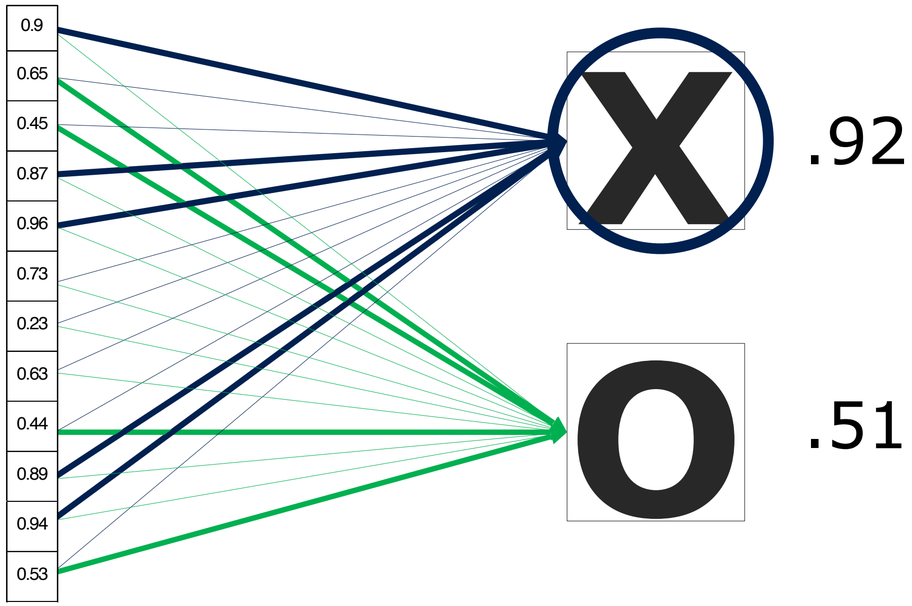
Future values vote on X or O



Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Fully connected layer

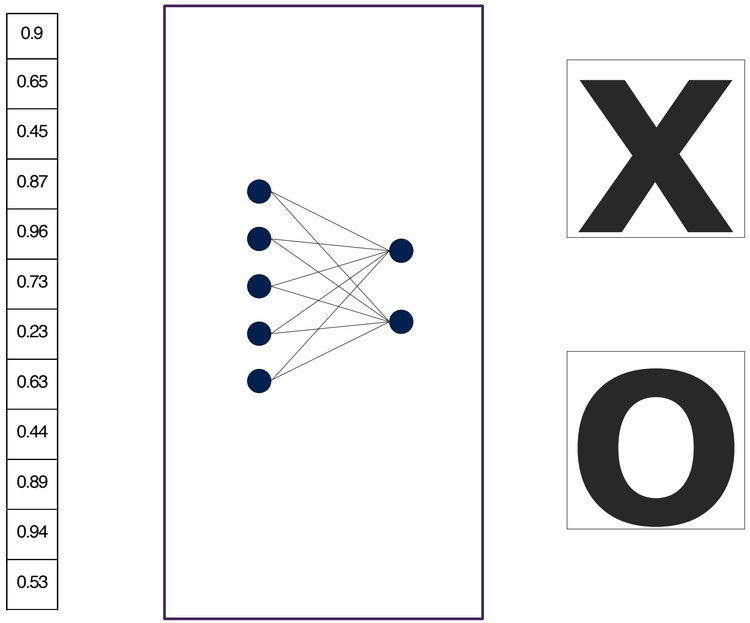
Future values vote on X or O



Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Fully connected layer

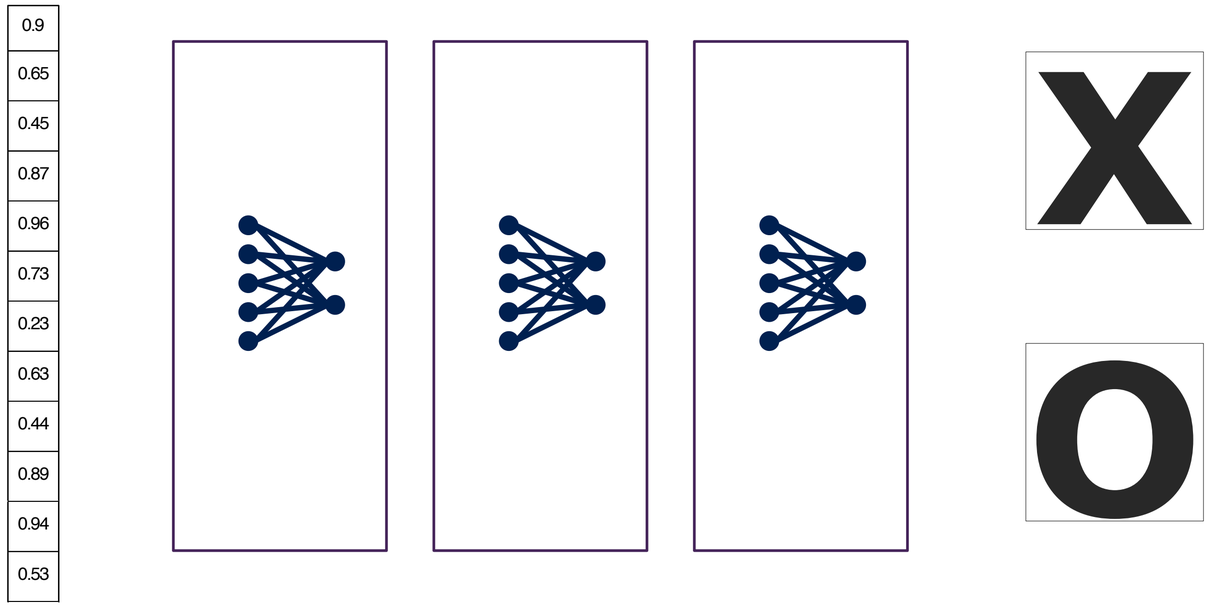
A list of feature values becomes a list of votes.



Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Fully connected layer

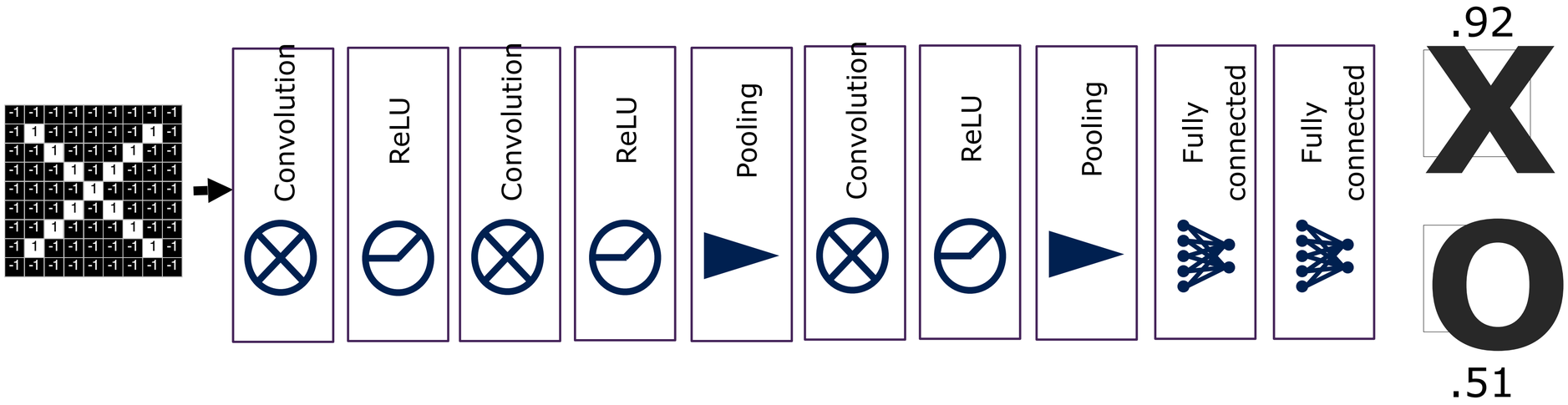
These can also be stacked.



Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html

Putting it all together

A set of pixels becomes a set of votes.



Brandon Rohrer https://brohrer.github.io/how_convolutional_neural_networks_work.html